

Reactor capital cost governs small modular reactor cogeneration for data-center power and cooling

Honglin Li^{1,3}, Buxin She³, Jie Zhang^{1,2*}

¹Department of Mechanical Engineering, The University of Texas at Dallas, Richardson, 75080, TX, USA.

²Department of Electrical and Computer Engineering, The University of Texas at Dallas, Richardson, 75080, TX, USA.

³Department of Electrical and Computer Engineering, Kansas State University, Manhattan, 66506, KS, USA.

*Corresponding author(s). E-mail(s): jiezhang@utdallas.edu;

Abstract

Data centers are becoming large, continuous electricity loads, but the value of colocating them with small modular reactors that also recover heat for cooling remains uncertain under current market and policy conditions. Here we use an hourly techno-economic co-optimization of reactor operation, steam extraction, absorption cooling and grid exchange to evaluate a 200 MW_e data center in ERCOT across market, cost and policy sensitivities. At 2023 mid-range reactor capital, the nuclear configurations cost 146–149% more than grid supply even with the nuclear production tax credit, whereas nth-of-a-kind reactor capital makes them 22–25% cheaper without a carbon price. Absorption cooling supplies 94% of annual cooling but changes cost by only \$2.4 M yr^{−1}, showing that reactor capital, not heat recovery, sets the viability boundary.

Keywords: Small modular reactor, Absorption cooling, Data center, Techno-economic analysis, Mixed-integer linear programming, Thermal integration

1 Introduction

Data centers are emerging as a major driver of electricity-demand growth after a decade in which efficiency gains moderated their global energy use¹. U.S. data-center electricity consumption nearly doubled between 2019 and 2023 and is projected to

5 reach 6.7–12% of total U.S. electricity by 2028²; globally, electricity use from data
6 centers is projected to roughly double to 945 TWh by 2030^{3,4}. In the Electric Reliability
7 Council of Texas (ERCOT), large-load interconnection requests already exceed the
8 scale of conventional planning margins^{5,6}. This growth creates a supply problem that
9 is not only about annual energy: hyperscale campuses are continuous loads, require
10 high availability, and carry cooling demands that rise sharply during hot and humid
11 hours^{7,8}.

12 Small modular reactors (SMRs) have therefore re-entered data-center procurement
13 discussions as firm, low-carbon generators whose unit sizes are closer to a hyperscale
14 campus than conventional gigawatt-scale nuclear plants^{9–12}. Yet the economic bound-
15 ary remains unsettled. Prior techno-economic assessments show that SMRs can be
16 lower-risk to build than large reactors but often remain above grid supply without a
17 carbon constraint¹². Capital-cost estimates also span a wide first-of-a-kind (FOAK)
18 to nth-of-a-kind (NOAK) range, with learning, standardization, financing and project
19 delivery determining whether reactors approach published low-cost trajectories^{11,13,14}.
20 Current U.S. policy adds a second uncertainty: the Section 45U nuclear production tax
21 credit (PTC) can materially change the cost of operating a qualifying nuclear plant,
22 but only if the pre-credit gap is small enough for the credit to close.

23 Cooling-side thermal integration offers a different route to value. Data-center cool-
24 ing can account for 20–40% of facility energy use, and absorption chillers can convert
25 heat into chilled water when the heat source is sufficiently hot and available^{15–17}.
26 Reactor steam cycles can provide the driving temperatures required by double-
27 effect lithium-bromide absorption chillers more readily than server exhaust, but the
28 value of this integration depends on chiller capital cost, extracted-steam opportunity
29 cost, wet-bulb-driven de-rating and backup electric cooling during crystallization-risk
30 hours^{18–20}. Treating heat recovery as an automatic benefit therefore risks overstating
31 the value of nuclear cogeneration for an efficiently cooled campus.

32 The remaining gap is a boundary problem. Existing studies rarely combine hourly
33 cooling physics, multi-year electricity and gas prices, reactor capital-cost trajectories,
34 production-tax-credit policy and carbon accounting in one co-optimization. One-
35 dimensional sensitivities also make it difficult to tell whether a proposed site or
36 contracting structure lies in a competitive region. The needed result is not a single
37 levelized cost but a multi-axis map: when does a colocated reactor beat grid procure-
38 ment, when does absorption cooling actually add value, and which assumptions move
39 the boundary?

40 Here we present a techno-economic boundary analysis of a 200 MW_e data center
41 colocated with a BWRX-300 SMR and a double-effect absorption chiller driven
42 by cascaded high-pressure to low-pressure turbine extraction. A mixed-integer linear
43 program co-optimizes power supply, cooling and grid exchange at hourly dispatch res-
44 olution across 2022–2024 ERCOT price, IT-load and Houston wet-bulb traces, with
45 daily Henry Hub spot prices plus basis applied to hourly gas-dispatch intervals and
46 optional battery storage evaluated as a sensitivity. We compare grid supply, reactor
47 supply without heat recovery, reactor supply with absorption cooling, and an on-site
48 natural gas combined cycle (NGCC) across 100 model runs spanning data-center scale,
49 cooling efficiency, market year, battery deployment, reactor and absorption capital

cost, carbon price and weighted-average cost of capital (WACC); the case definitions, optimization formulation and scenario ledger are specified in Methods and the Supplementary Information. We find that nuclear cogeneration is a conditional strategy: at mid-range reactor capital it remains far above grid cost, but at NOAK reactor capital the PTC opens a viable region with no carbon price, while absorption cooling supplies most cooling yet does not set the economic boundary.

2 Results

2.1 A boundary analysis separates supply, heat recovery and policy effects

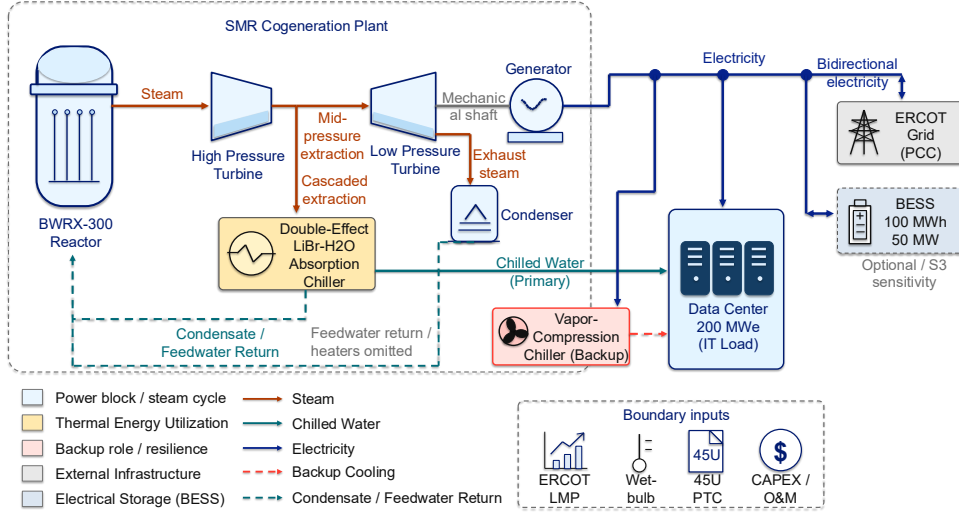


Fig. 1 Cascaded steam-extraction mechanism for SMR-powered data-center cooling. Steam from the BWRX-300 power block expands through the high-pressure turbine, a mid-pressure extraction stream drives a double-effect LiBr-H₂O absorption chiller, and the remaining steam expands through the low-pressure turbine to produce electricity. Chilled water supplies the data center, vapor-compression cooling provides backup during absorption limits, and the grid and battery interfaces balance electricity. Complete case definitions, optimization equations and scenario groups are provided in Methods and the Supplementary Information. SMR, small modular reactor; LiBr, lithium bromide; ERCOT, Electric Reliability Council of Texas; PCC, point of common coupling; BESS, battery energy storage system; CAPEX, overnight capital expenditure; O&M, operations and maintenance.

The analysis is organized around a single competitiveness metric, the grid-cost margin, defined as one minus the ratio of each configuration's total annualized cost (TAC) to the matching grid-only data-center cost. Positive values mean that an on-site option beats grid procurement; negative values mean that it does not. This framing matters

because the reactor, cooling system and policy support can move the margin in different directions. Reactor capital and financing determine the annualized fixed-cost burden, absorption cooling changes both chiller electricity and turbine output, grid exchange determines merchant-export value, and the Section 45U PTC offsets eligible nuclear generation.

Figure 1 presents the fully integrated Case 2 configuration as the organizing system schematic, with reactor power conversion, cascaded steam extraction, absorption cooling, vapor-compression backup, grid exchange and optional storage shown in one architecture. The four model configurations are then defined as supply and cooling variants of this architecture in Methods and Table 1, the hourly optimization workflow is specified by the formulation in Methods, and the scenario envelope is given in the scenario-design subsection and Supplementary Table S9. The Results first establish the baseline economic gap, then ask whether operations, cooling-side value, reactor capital, financing, market year or carbon policy can close it.

2.2 Mid-range reactor capital keeps nuclear options above grid supply

At the 2023 National Laboratory of the Rockies (NLR) Annual Technology Baseline mid-range (ATB-Mid) reactor-capital baseline, the nuclear configurations are not close to grid parity. The grid-supplied data center (Case 0) has a TAC of \$78.1 M yr⁻¹, of which 93% is the electricity bill. The two reactor configurations cost \$192.2 and \$194.6 M yr⁻¹ without heat recovery (Case 1) and with absorption cooling (Case 2), respectively, even after the Section 45U PTC returns \$26.4 and \$25.8 M yr⁻¹ (Figure 2a). Reactor capital recovery dominates both totals. Grid-export revenue and the PTC offset part of the fixed-cost burden, but not enough to prevent grid-cost margins of -146% and -149%.

The only baseline configuration that beats grid procurement is the on-site NGCC plant (Case 3), at \$55.7 M yr⁻¹ and a +29% grid-cost margin. This result is tied to the 2023 fuel-price regime: delivered natural gas averaged \$2.63 MMBtu⁻¹, while ERCOT day-ahead prices remained high enough for the grid-only bill to exceed on-site fuel cost. Because Case 3 is islanded, it cannot import during the roughly 42% of hours when the day-ahead price falls below its running cost, so the comparison is an annual cost result rather than an hour-by-hour arbitrage result. Adding absorption cooling to the nuclear plant changes cost by only \$2.4 M yr⁻¹ relative to the reactor-only case, showing already at the baseline that heat recovery is not the primary economic lever.

The BWRX-300 unit-size constraint explains part, but not most, of the baseline nuclear penalty. The smallest commercial unit is 270 MW_e net at the assumed capacity factor, about 2.9 times the data center's 94.7 MW_e average information-technology (IT) load, so Case 2 net-exports about 1.43 TWh yr⁻¹ to ERCOT, with 2.16 TWh yr⁻¹ of gross sales against 0.73 TWh yr⁻¹ of grid imports. Scaling the campus to 2.5× the baseline load brings the nuclear site close to import-export balance, but the matched Case 2 margin is still -62%; at 3.0× the site becomes a net importer and the margin improves only to -52% (Supplementary Figure S3). Unit-size matching therefore improves utilization intensity, reducing the Case 2 excess cost from \$140 to \$49 MWh_{IT}⁻¹ across the sweep, but it does not remove the ATB-Mid reactor-capital gap.

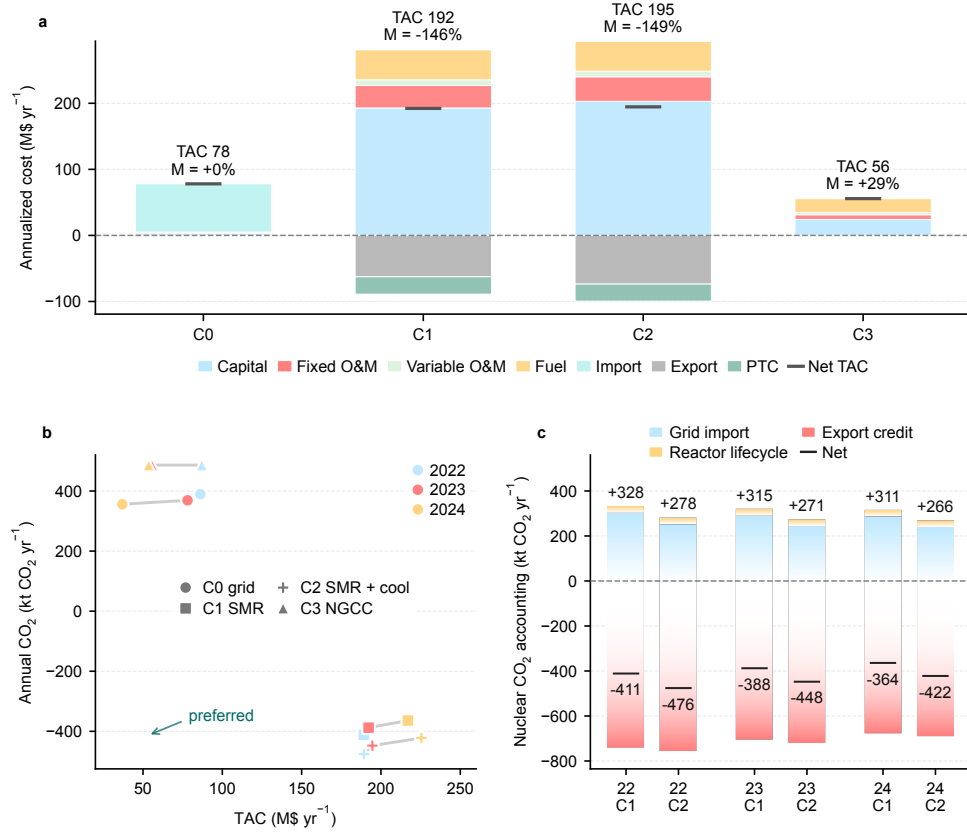


Fig. 2 Baseline economics of the four configurations at the 2023 ATB-Mid baseline. ATB-Mid denotes the mid-range National Laboratory of the Rockies Annual Technology Baseline reactor-capital case; SMR, small modular reactor; NGCC, natural gas combined cycle; PUE, power usage effectiveness; WACC, weighted-average cost of capital; O&M, operations and maintenance; PTC, production tax credit; TAC, total annualized cost; ERCOT, Electric Reliability Council of Texas. Baseline assumptions are full-load PUE 1.35 and WACC 6.7%. (a) TAC decomposition: each bar stacks capital recovery, fixed and variable O&M, fuel and grid imports; bars below zero are grid-export revenues and, for the nuclear cases, the Section 45U PTC; the horizontal mark is the net TAC and M is the grid-cost margin against Case 0. (b) Cost-carbon trade-off across ERCOT years 2022, 2023 and 2024: marker shape denotes case, marker color denotes year, and the arrow indicates the Pareto-preferred direction. (c) Carbon-accounting components for the two nuclear configurations: grid-import and reactor-lifecycle emissions are offset by grid-export credits, and the horizontal mark reports net annual CO₂.

The baseline cost ranking is mirrored in the cost-carbon plane (Figure 2b,c). The nuclear cases occupy the low-carbon region after reactor-lifecycle emissions and grid-export credits are accounted for using the ERCOT hourly average carbon intensity, while the grid-only case anchors the low-cost region in 2023 and 2024. Case 3 trades higher emissions for lower cost under 2023 gas prices. Figure 2c shows that export credits more than offset grid-import and reactor-lifecycle emissions in Cases 1 and

2, but low carbon intensity alone does not make either reactor configuration cost-competitive at zero carbon price and mid-range reactor capital.

Secondary endpoints lead to the same boundary interpretation rather than a different ranking (Supplementary Tables S5 and S6). At the baseline, Case 2 has a levelized electricity cost of \$234.5 MWh_e⁻¹, a carbon-abatement cost of \$142.7 tCO₂⁻¹ relative to the grid-only case, an energy payback time of 0.53 yr and a total water footprint of 8,491 L MWh_e⁻¹. These metrics show why the low-carbon reactor cases remain expensive at ATB-Mid capital: their net average-emissions footprint is negative under the export-credit convention, but the capital-driven cost and water-footprint penalties remain large.

2.3 Absorption cooling supplies most cooling but adds little value

The hourly dispatch of Case 2 shows how the absorption chiller integrates with the reactor under winter and summer operating conditions (Figure 3a,b). In the winter week, the BWRX-300 runs near nameplate thermal output, consistent with its 92% annual capacity-factor constraint. The high-pressure to low-pressure turbine train produces close to its full ~ 270 MW_e net output through most of the window, turning down to about 125 MW_e during the low-price hours. Chilled-water demand tracks the 25–142 MW_e winter IT load through $Q_{\text{cool}}^{\text{dem}} = P_{\text{IT}}/\eta_{\text{ch}}$, and the extraction tap supplies most of it as steam, with the VCC backup contributing only on the highest-demand hours. In the July week, wet-bulb temperature climbs toward the 26 °C design anchor on the warm afternoons, derating the absorption coefficient of performance (COP) toward its 1.10 design value. The week stays below the 27 °C crystallization gate (32 °C cooling-water inlet). The higher summer cooling load nonetheless exceeds what the derated absorber can supply from steam, so the VCC backup tops up to about 30 MW_c on the warmest afternoons. Across the full year the absorption chiller delivers 862 GWh_c and the VCC backup 60 GWh_c, the latter spread over roughly 2,870 hours of partial top-up that also cover the 5% maintenance derating of the absorber (Supplementary Information). The crystallization gate itself is rare: the availability indicator in Equation 8 forces the absorber fully offline for about 115 hours, concentrated in late June through August (annual peak wet-bulb 28 °C), during which the backup carries the entire cooling load of those hours (13.6 GWh_c).

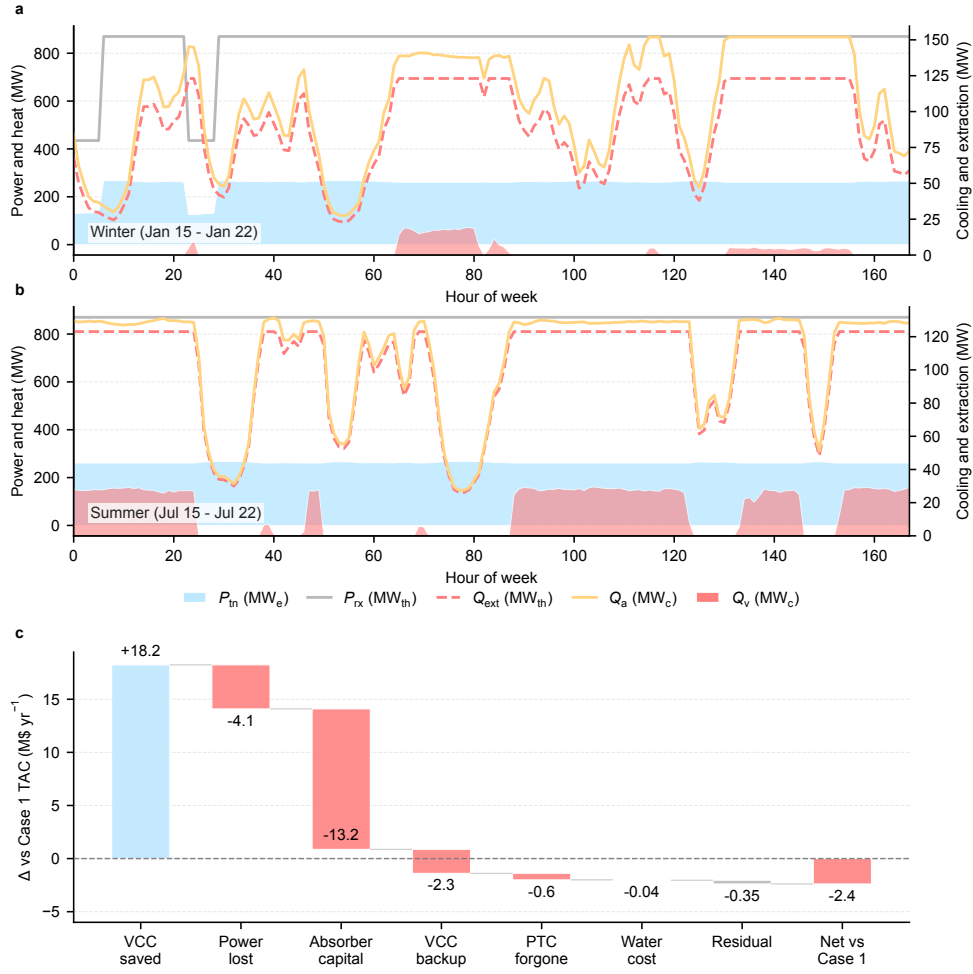


Fig. 3 Cogeneration operation and the marginal value of absorption cooling for Case 2. VCC denotes vapor-compression chiller; TAC, total annualized cost; PTC, production tax credit. (a) Hourly dispatch over a winter week (15–22 January) and (b) a summer week (15–22 July): reactor thermal output P_{rx} and turbine net electric output P_{tn} are read on the left axis; mid-pressure extraction Q_{ext} , absorption cooling Q_a , and VCC backup cooling Q_v on the right axis. In both plotted weeks the wet-bulb stays below the 27 °C crystallization gate, so the absorption chiller carries most of the cooling load; the VCC backup tops up on the warmest summer afternoons, when the derated absorber cannot meet demand from steam, and takes over entirely during the roughly 115 crystallization-gate hours yr⁻¹ elsewhere in the record. (c) Waterfall decomposition of the marginal annual value of adding the absorption chiller to a nuclear-only plant (Case 2 against Case 1): positive bars reduce Case 2 TAC, negative bars increase it, VCC backup covers steam-limited top-up and crystallization-gate fallback, PTC forgone is the production tax credit lost as steam diverts from the turbine, and the residual bar reconciles the component sum with the measured \$2.4 M yr⁻¹ difference.

The waterfall in Figure 3c decomposes the $\$2.4 \text{ M yr}^{-1}$ gap between Case 2 and Case 1 into the marginal components of adding absorption cooling to the nuclear-only plant. The chiller displaces $\$18.2 \text{ M yr}^{-1}$ of gross vapor-compression electricity at the ERCOT 2023 hourly import price. Against this credit stand, in annual terms, $\$13.2 \text{ M}$ for absorption capital plus fixed operations and maintenance, $\$4.1 \text{ M}$ for the Willans-line electricity opportunity cost of extracted steam (the marginal turbine-output loss caused by steam diversion), $\$2.3 \text{ M}$ for VCC backup and top-up, and $\$0.6 \text{ M}$ for the reduction in the production tax credit, since diverting steam to the absorber lowers the creditable turbine output; the monetized direct-site-water increment is only $\$0.04 \text{ M yr}^{-1}$. A small $-\$0.3 \text{ M yr}^{-1}$ residual reconciles the component sum with the measured TAC difference. Credit and cost nearly cancel, so the chiller is close to break-even, and the residual penalty is a capital-cost problem rather than one of operating physics or water.

With the heat-rejection load fixed at $P_{\text{IT}}/\eta_{\text{ch}}$, the cooling-side lever is the efficiency of the competing electric chiller rather than the size of the duty. The cooling-efficiency sweep (Supplementary Figure S2) varies the chiller coefficient of performance so that the reported power usage effectiveness (PUE) spans $\{1.10, 1.30, 1.50\}$ for Cases 1 and 2, with all other parameters at the 2023 ATB-Mid baseline. The absorber is not forced to run whenever cooling demand exists: at PUE 1.10, highly efficient electric cooling makes steam diversion unattractive for much of the year and the absorption chiller delivers only $126 \text{ GWh}_c \text{ yr}^{-1}$. At PUE 1.30 and 1.50, by contrast, it supplies 861 and $863 \text{ GWh}_c \text{ yr}^{-1}$, respectively. What changes most across the sweep is therefore both absorber dispatch at the most efficient electric-cooling endpoint and the value of the electric cooling displaced when the absorber runs. Net cooling-electricity saving after VCC backup and top-up rises from $\$1.9$ to $\$13.8$ and $\$23.0 \text{ M yr}^{-1}$ as PUE increases from 1.10 to 1.30 and 1.50. As a result, Case 2 costs $\$13.0 \text{ M yr}^{-1}$ more than Case 1 when the electric chiller is highly efficient, narrows to a $\$4.6 \text{ M yr}^{-1}$ penalty at PUE 1.30, and becomes a $\$4.6 \text{ M yr}^{-1}$ saving at PUE 1.50. The main-results baseline uses a COP-3.2 electric chiller at a full-load PUE of 1.35, so its $\$2.4 \text{ M yr}^{-1}$ Case 2-minus-Case 1 gap sits just below the break-even point near PUE 1.40.

2.4 Viability opens only along the low reactor-capital frontier

The ERCOT price regime during the year of operation is the largest single market sensitivity (Figure 4a). The 2022 cluster captures the high natural-gas price regime: delivered natural gas (Henry Hub plus Houston Ship Channel basis) averaged $\$6.53 \text{ MMBtu}^{-1}$, and the NGCC margin falls to -1% , essentially parity with the grid, because the grid clearing price and the on-site fuel cost moved together. In 2023, low gas prices ($\$2.63 \text{ MMBtu}^{-1}$) combined with high ERCOT prices made Case 3 the only on-site option that beat grid supply (a $+29\%$ margin). The 2024 cluster shows the deepest negative margin for every on-site option because ERCOT LMPs settled at a multi-year low, an annual mean of about $\$28 \text{ MWh}_e^{-1}$, making grid procurement comparatively inexpensive. Across the three years, no on-site option beats grid supply in 2024 or 2022, and only NGCC beats it in 2023. The year regime therefore sets the market boundary for the on-site gas option: a $\pm 50\%$ swing in Henry Hub gas price or ERCOT LMP can overturn the ranking between grid and NGCC supply. Nuclear, by

contrast, stays far above grid cost in all three years, so its ranking is governed by the capital-cost rather than the market axis.

The battery energy storage system (BESS) axis tests whether a 100 MWh, 50 MW lithium-ion battery changes the competitiveness frontier. It lowers TAC by \$0.63 M yr⁻¹ in each nuclear case, improving the margin by only 0.8 percentage point and never altering the configuration ranking (see Supplementary Information). Because this effect is much smaller than the market, capital-cost and financing axes, it is retained as a secondary boundary rather than as a main result.

SMR overnight capital cost is the dominant technology-cost axis (Figure 4b). At FOAK (\$14,700 kW_e⁻¹) the margin is -372% (Case 1) and -375% (Case 2): reactor capital outweighs all other costs by roughly a factor of five, and the production tax credit is far too small to matter. At ATB-Mid the margin settles at -146% (Case 1) and -149% (Case 2), so even with the credit, mid-range capital leaves the nuclear cases about 2.5 times more expensive than grid supply. At NOAK (\$2,250 kW_e⁻¹) the picture inverts: the margin rises to +25% for Case 1 and +22% for Case 2, so both nuclear configurations are cheaper than grid procurement with no carbon price applied. The credit is decisive precisely here, because the pre-credit gap to grid at NOAK capital is small, on the order of 10%, and the \$26.4 M yr⁻¹ Case 1 credit more than covers it, whereas the same credit barely dents the far larger FOAK and ATB-Mid gaps. Coupling the NOAK reactor with the absorption chiller (Case 2 NOAK) is marginally less competitive than Case 1 NOAK, because the absorption capital still slightly outweighs the cooling-side savings, but both stay comfortably below grid cost.

Financing sensitivity. The financing assumption changes the same capital-dominated boundary rather than creating a separate operating mechanism (Figure 4c). Sweeping the weighted-average cost of capital between 5% and 10% for Case 2 while holding absorption capital, reactor capital and operating parameters fixed moves TAC from \$168 to \$251 M yr⁻¹, an \$83 M yr⁻¹ range. The implied grid-cost margin ranges from -115% at 5% WACC to -221% at 10% WACC. By comparison, the full range of group G4 (reactor capital, FOAK to NOAK) moves the Case 2 margin between -375% and +22%, crossing the parity line. Financing materially affects the high-cost FOAK-ATB region but matters less than the reactor capital-cost axis once the trajectory reaches NOAK. Policy or contracting mechanisms that lower the effective cost of capital for SMR projects, such as long-term offtake agreements, sovereign loan guarantees, or accelerated depreciation, can compress but not eliminate the gap to grid supply.

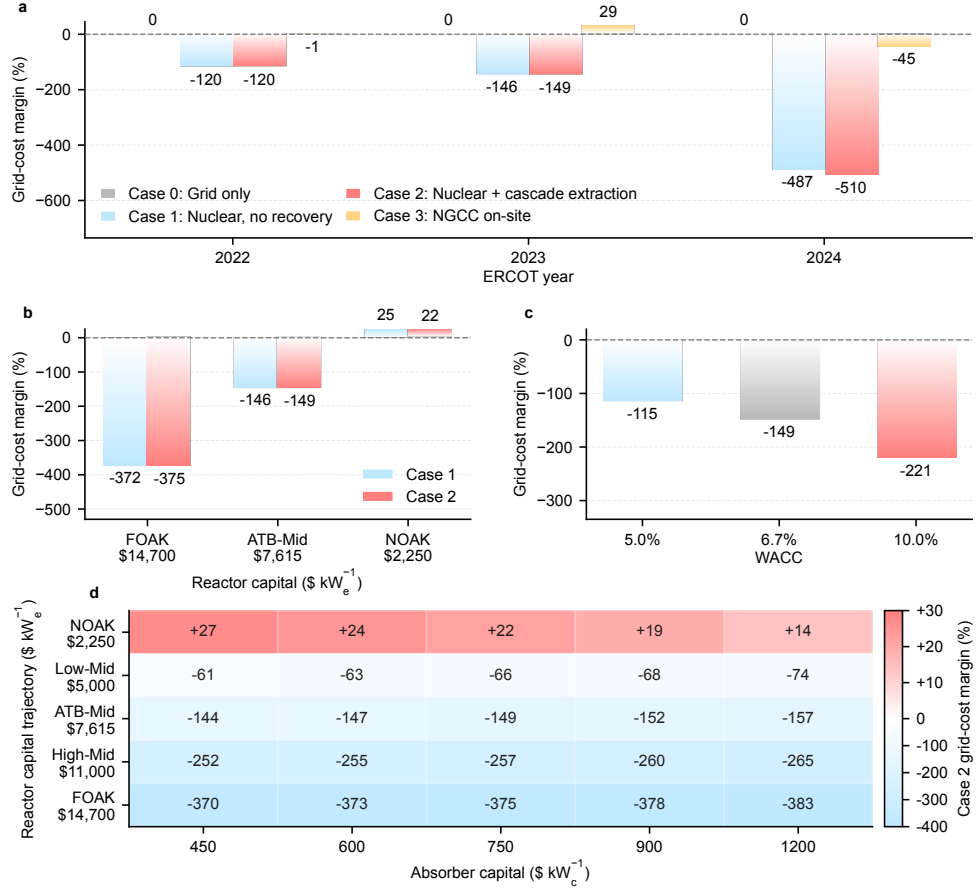


Fig. 4 Market, reactor-capital, financing and joint capital-cost boundary atlas. ERCOT denotes Electric Reliability Council of Texas; NLR ATB, National Laboratory of the Rockies Annual Technology Baseline; WACC, weighted-average cost of capital; SMR, small modular reactor; FOAK, first-of-a-kind; NOAK, nth-of-a-kind. (a) Grid-cost margin for the four configurations across ERCOT years 2022, 2023 and 2024. (b) Grid-cost margin for Cases 1 and 2 across the NLR ATB 2024 BWRX-300 capital-cost trajectories. (c) Case 2 grid-cost margin across the WACC sweep. (d) Joint SMR and absorption-chiller capital-cost viability frontier for Case 2; cell values are grid-cost margin against the 2023 grid-only baseline. Bars or cells below zero indicate configurations that are more expensive than the grid-only baseline.

224 The joint capital-cost frontier in Figure 4d turns the preceding one-axis sensitivities
 225 into the procurement-relevant capital-cost test. Both axes sweep simultaneously: SMR
 226 capital cost runs from NOAK to FOAK on the vertical axis ($\$2,250\text{--}14,700 \text{ kW}_e^{-1}$),
 227 and absorption-chiller capital cost runs from Bare-Low to Turnkey-High on the hori-
 228 zontal axis ($\$450\text{--}1,200 \text{ kW}_c^{-1}$). Every cell in the 5×5 grid is solved at 2023 ATB-Mid
 229 market conditions, full-load PUE 1.35, WACC 6.7%, and zero carbon price, with the
 230 production tax credit applied. A viable region now opens along the NOAK edge:

every cell in the NOAK row reaches a positive grid-cost margin, from +27% with Bare-Low absorption to +14% with Turnkey-High, so the margin = 0 contour enters the published rectangle. The most favorable corner (NOAK reactor with a Bare-Low absorption chiller at $\$450 \text{ kW}_e^{-1}$) lands at +27%, and the least favorable (FOAK reactor with Turnkey-High absorption) at -383%. The frontier is governed almost entirely by the reactor axis: absorption capital shifts the margin by at most 13 percentage points across its full published range, whereas the internal Low-Mid reactor row at $\$5,000 \text{ kW}_e^{-1}$ already drops the Case 2 margin to -61 to -74% across the absorption-cost sweep. Reaching roughly nth-of-a-kind reactor capital is therefore the binding condition. With the production tax credit it makes colocated SMR with absorption cooling cheaper than grid supply at 2023 ERCOT prices and zero carbon cost, and no tested capital trajectory above the NOAK row is competitive.

2.5 Carbon pricing is a contingent policy pathway

A carbon price provides a less robust policy pathway to nuclear competitiveness, distinct from the NOAK capital route just described. Holding capital at the ATB-Mid baseline, the policy axis is the carbon price (Figure 5a). All four cases have net annual CO₂ footprints that are nearly fixed by dispatch (Case 0 +369, Case 3 +486, Case 1 -388 and Case 2 -448 kt CO₂ yr⁻¹), so each TAC-carbon relationship is essentially linear over the plotted range. The nuclear footprints barely deepen as carbon is valued, because the production tax credit already drives the reactor to export close to its full surplus, leaving little additional export to unlock.

The grid-only Case 0 increases by $\$0.37 \text{ M yr}^{-1}$ for each $\$1 \text{ tCO}_2^{-1}$ rise in carbon price because the ERCOT grid mix is carbon-intensive. The two nuclear cases decrease by about $\$0.39\text{--}0.45 \text{ M yr}^{-1}$ under the same price increment, because their grid exports displace carbon-intensive generation and receive a corresponding credit. Case 3 NGCC increases by $\$0.49 \text{ M yr}^{-1}$ for each $\$1 \text{ tCO}_2^{-1}$ rise, carrying the full direct emissions on the buyer side. As the carbon price rises, the two on-site options cross the grid-only baseline in opposite directions. At $\$100 \text{ tCO}_2^{-1}$, the highest carbon price explicitly solved in group G6, Case 2 still costs 30% more than Case 0 and Case 1 costs 33% more; the reported nuclear crossover range of $\$142\text{--}\151 tCO_2^{-1} is therefore an extrapolation beyond the $\$0\text{--}\100 tCO_2^{-1} solved range. NGCC starts below grid and rises above it near an extrapolated $\$191 \text{ tCO}_2^{-1}$. This carbon-pricing pathway is independent of the NOAK capital route: at NOAK capital the nuclear cases already sit below grid cost at zero carbon price.

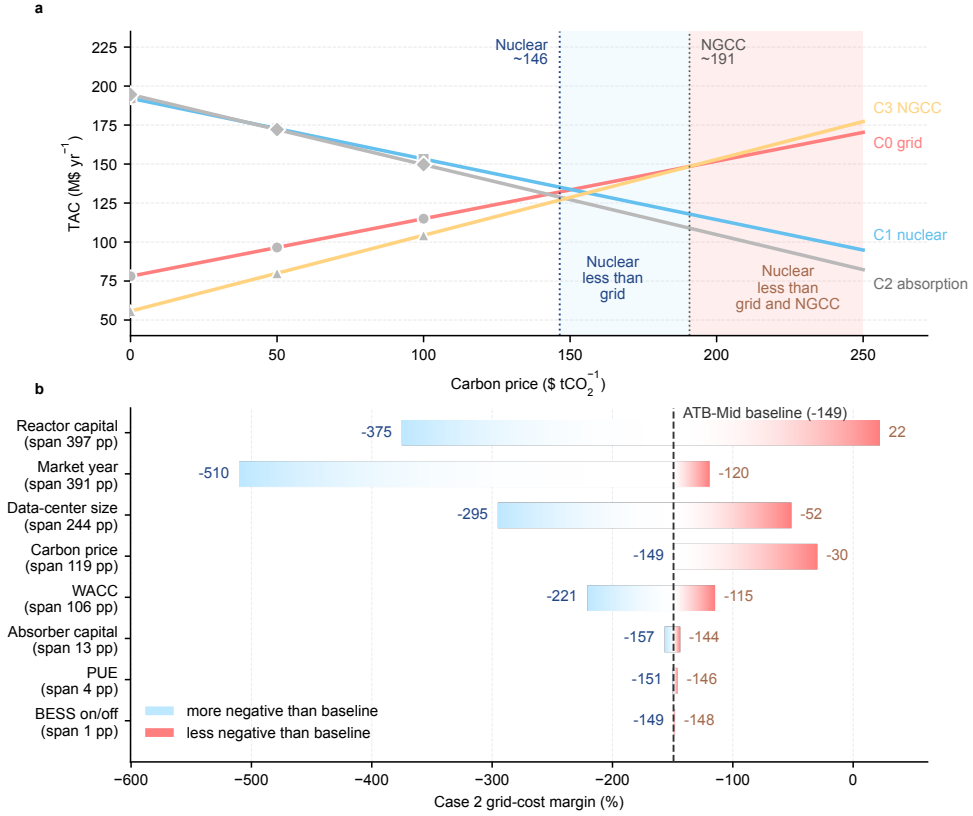


Fig. 5 Policy crossover and global sensitivity ranking. TAC denotes total annualized cost; NGCC, natural gas combined cycle; ATB-Mid, mid-range National Laboratory of the Rockies Annual Technology Baseline reactor-capital case; SMR, small modular reactor; BESS, battery energy storage system; PUE, power usage effectiveness; WACC, weighted-average cost of capital; CAPEX, overnight capital expenditure. (a) TAC extrapolated against carbon price for the four configurations. Vertical dotted lines mark the nuclear and NGCC crossovers against Case 0; the blue region marks where the nuclear cases begin to beat grid supply, and the pink region marks the higher-carbon-price range where they are below both grid and NGCC. (b) Ranked sensitivity of the Case 2 grid-cost margin to each boundary axis, anchored at the 2023 ATB-Mid baseline (−149%).

The cogeneration cooling earns only a small, indirect carbon benefit: meeting cooling from steam rather than electricity lets Case 2 both export more and import less than Case 1, giving it a marginally deeper net footprint (−448 versus −388 kt CO₂ yr^{−1}) and a slightly earlier crossover, but it does not create a separate credit mechanism. Both crossings depend on crediting the reactor’s merchant exports at the ERCOT hourly average carbon intensity: the nuclear cases reach a net-negative footprint only because the ~1.43 TWh yr^{−1} of net exported electricity is credited as displaced grid generation. Because export-credit retention is not varied in the solved

G6 sweep, this accounting choice is treated as a limitation rather than as an additional solved boundary.

The ranked sensitivity panel in Figure 5b summarizes why the policy route is secondary. Around the Case 2 ATB-Mid baseline (-149%), the full one-axis span is 397 percentage points for SMR capital cost (-375 to $+22\%$) and 391 percentage points for market year (-510 to -120%). Data-center size contributes a 244-percentage-point span, while carbon price and WACC contribute 119 and 106 percentage points, respectively. Absorption capital, PUE and BESS deployment are much smaller axes at 13, 4 and 1 percentage points. Thus the global ranking in the solved envelope is consistent with the one-dimensional results: reactor capital and market regime determine the boundary, carbon pricing can move the ATB-Mid case toward parity, and cooling-side refinements do not overturn the capital-cost ordering.

3 Discussion and conclusions

Colocated small modular reactors with absorption cooling are a conditional, not categorical, low-carbon strategy for hyperscale data centers. Across the four configurations and 100 model runs, the consistent boundary is not whether heat recovery can technically serve the cooling load, but whether reactor capital cost has fallen far enough for the Section 45U production tax credit to close the remaining gap to grid procurement. The central conclusion is therefore that nuclear data-center cogeneration becomes viable only when reactor capital reaches the low-cost learning trajectory; heat recovery and storage refine the operating case but do not set the boundary.

The mechanism is a fixed-cost threshold rather than an operating-efficiency threshold. Sweeping reactor and absorption-chiller capital together, the entire nth-of-a-kind reactor row is competitive, from $+27\%$ with a bare-low absorption chiller to $+14\%$ with a turnkey-high one, whereas the internal Low-Mid reactor row at $\$5,000 \text{ kW}_e^{-1}$ is already 61–74% above grid cost. Absorption capital changes the margin by at most 13 percentage points over its published range. By contrast, the reactor-capital trajectory moves the margin by hundreds of percentage points, so the production tax credit is decisive only once the pre-credit capital gap is already small enough to close.

This finding changes how the data-center nuclear question should be framed. Unit-size matching removes an important distortion but does not create parity: scaling the load to $2.5\times$ brings the reactor close to import-export balance, yet the cogeneration plant remains 62% more expensive than the matched grid baseline. Market year mainly determines whether the on-site gas comparator is temporarily attractive; it does not rescue mid-range nuclear capital. Financing helps by lowering annualized fixed cost, but the 5–10% WACC sweep still leaves Case 2 far above grid supply at ATB-Mid capital. These results suggest that procurement discussions should treat load matching, market regime and financing as necessary supporting conditions rather than substitutes for reactor cost learning.

Carbon pricing is a policy pathway but a less robust one. Holding reactor capital at the mid-range baseline, Case 2 remains about 30% above the grid-only baseline even at the highest solved carbon price of $\$100 \text{ tCO}_2^{-1}$, so the near- $\$150 \text{ tCO}_2^{-1}$ crossover is extrapolated beyond the solved G6 range. The low-capital route is therefore more

decision-relevant because it does not depend on extending the carbon-price fit or on a particular export-credit accounting convention.

Against this capital-dominated boundary, cooling integration should be interpreted as a reliability and energy-management design choice rather than the main economic source of value. The absorption chiller supplies $862 \text{ GWh}_c \text{ yr}^{-1}$, about 94% of annual cooling, but it raises cost by $\$2.4 \text{ M yr}^{-1}$ at the baseline because its capital cost, the lost turbine output and a smaller production tax credit slightly exceed the avoided electric cooling cost (Section 2.3). The sign reverses when full-load PUE exceeds about 1.40 or when carbon is priced, so absorption can matter for inefficient campuses or carbon-constrained settings. For an efficient COP-3.2 electric chiller, however, thermal integration is near cost-neutral. This refines waste-heat studies that treat absorption as categorically beneficial and reactor techno-economic studies that omit cooling-side physics.

Several limitations bound the quantitative margins without overturning the central ordering. The load-scaling sweep uses one hyperscale profile, so campuses dominated by different artificial-intelligence workloads or located in different climates could shift the cooling value. The carbon-policy route uses hourly average rather than marginal emission factors, so a merit-order marginal series would better estimate displaced emissions from nuclear exports. Demand charges and ancillary-service revenues are not modelled; adding delivered retail tariffs or market-service revenues would change the grid and export economics. The gas comparator omits forced outages, redundant generation and an outage-covering grid tie, so five-nines availability requirements would erode its baseline advantage. The production tax credit is applied at full prevailing-wage value to net nuclear generation under an assumed power-purchase-agreement structure and held at nominal in-window values, so eligibility limits, credit expiry or inflation-indexing details would alter the low-capital result. The reactor is also treated as the smallest deployable commercial unit rather than an optimized multi-module buildout.

For procurement, the result is a short list of conditions that must coincide: reactor capital on the nth-of-a-kind trajectory, production-tax-credit eligibility, a load large enough to use the smallest commercial reactor and a market regime in which grid procurement is not unusually cheap. Where those conditions hold, small modular reactor cogeneration can undercut grid supply without a carbon price, and absorption cooling can be chosen for operational reasons rather than as the source of the saving. Where reactor capital remains at first-of-a-kind or mid-range levels, neither thermal integration nor a plausible carbon price is enough. The broader contribution of the study is the boundary map itself: it provides a way to test nuclear data-center proposals against region-specific tariffs, financing, cooling climates and carbon-accounting rules before treating firm low-carbon supply as an economic proposition.

4 Methods

4.1 System configurations

The analysis compares four configurations that supply electricity and chilled water to a single 200 MW_e rated hyperscale data center in Houston, Texas. Each configuration

Table 1 System configurations modeled in this study. ✓ denotes that the block is enabled; – that it is disabled. SMR denotes small modular reactor; NGCC, natural gas combined cycle; ERCOT, Electric Reliability Council of Texas; PCC, point of common coupling; VCC, vapor-compression chiller; LiBr, lithium bromide; BESS, battery energy storage system.

		Case 0	Case 1	Case 2	Case 3
Generation	BWRX-300 SMR + steam turbine	–	✓	✓	–
	NGCC on-site, F-class	–	–	–	✓
	ERCOT grid (PCC, 300 MW)	✓ (import)	✓ (bi-dir.)	✓ (bi-dir.)	–
Cooling	VCC, water-cooled	✓ (primary)	✓ (primary)	✓ (backup)	✓ (primary)
	Double-effect LiBr absorber	–	–	✓	–
Storage	Li-ion BESS, 100 MWh, 50 MW	–	sensitivity	sensitivity	–

uses the same data-center load profile, vapor-compression chiller technology when required, and financial parameters; the configurations differ only in the electricity supply pathway and in whether the cooling chain uses recovered reactor heat. Figure 1 illustrates the cogeneration mechanism and the grid and storage interfaces for the thermal-integration case. Cases 0, 1 and 3 are reduced versions of this layout in which selected blocks are disabled. Table 1 summarizes the four configurations.

Case 0 represents the conventional grid-supplied data center. All electricity is purchased from ERCOT at the day-ahead locational marginal price, and all cooling is produced by a water-cooled centrifugal vapor-compression chiller with a Houston-derated coefficient of performance of 3.2. This case serves as the benchmark for the grid-cost margin reported throughout the paper. Case 1 replaces grid imports with on-site BWRX-300 generation while retaining the same VCC. Heat at the turbine condenser is rejected to the cooling tower and is not recovered. Bidirectional grid exchange is retained so that excess electricity can be sold to ERCOT during low-load hours and grid imports can cover residual demand during planned outages. Case 2 adds a 160 MW_{th} nominal double-effect LiBr–H₂O absorption unit fed by a mid-pressure extraction tap on the high-pressure (HP) turbine; effective delivered cooling is limited by steam availability, COP and the wet-bulb availability gate. The cascade keeps the low-pressure (LP) turbine in service, so diverted steam incurs a Willans penalty rather than the full turbine efficiency penalty. A 160 MW_c vapor-compression backup chiller remains installed, the same capacity as the primary chiller in the VCC-only cases, because it must carry the full cooling load whenever the absorption unit shuts down at the LiBr crystallization threshold. Case 3 replaces the SMR with a 200 MW_e on-site F-class natural gas combined cycle, retains the VCC, and disconnects from the grid to isolate the value of replacing grid procurement with a fossil-fueled on-site alternative.

The data-center IT load follows an annual hourly trace synthesized from a 200 MW_e hyperscale reference profile with a peak hour of 142.4 MW_e, an annual average of 94.7 MW_e, and an annual utilization of 0.67. The same trace is used in every case and

every scenario; the IT load is not a decision variable. We treat the full IT electrical draw as heat that the cooling system must reject, so the hourly cooling demand is

$$Q_{\text{cool}}^{\text{dem}}(t) = \frac{P_{\text{IT}}(t)}{\eta_{\text{ch}}}, \quad (1)$$

where $\eta_{\text{ch}} = 0.90$ accounts for the chilled-water distribution chain. Each scenario is labeled by its full-load vapor-compression PUE, $1 + 1/(\eta_{\text{ch}} \text{COP}_{\text{v}})$, which is 1.35 at the baseline COP of 3.2. The dispatch-realized annual PUE, $1 + P_{\text{cool}}/P_{\text{IT}}$, is slightly lower for the vapor-compression cases, near 1.33, because the part-load COP curve runs more efficiently than its full-load point. It falls to about 1.05 in Case 2, where steam-driven absorption supplies most of the cooling and leaves little electric cooling load. The cooling-efficiency sensitivity in Section 2.3 sweeps the chiller COP so that this full-load PUE label spans $\{1.10, 1.30, 1.50\}$, the range between high-efficiency and legacy hyperscale practice^{7,8}.

4.2 Optimization formulation

The four cases are formulated as separate dispatch problems on the same 8760-hour horizon. Cases 0 and 3 reduce to closed-form annual cost evaluations because the absence of storage and heat recovery makes the hourly decisions independent. Cases 1 and 2 are mixed-integer linear programs: the only integer structure is the special-ordered-set (SOS2) representation of the vapor-compression chiller part-load curve described below, which is non-convex and cannot be relaxed; no binary commitment variables are introduced, because round-trip losses in the battery block already make simultaneous charge and discharge unprofitable. The same accounting structure is applied to all four cases so that cross-case comparisons remain on a common basis.

Reactor and turbine block (Cases 1 and 2). The BWRX-300 reactor is modeled as a continuous thermal source with bounded ramping and an annual capacity-factor constraint. The reactor block enforces

$$\underline{P}_{\text{rx}} \leq P_{\text{rx}}(t) \leq \bar{P}_{\text{rx}}, \quad \forall t \in \mathcal{T}, \quad (2)$$

$$|P_{\text{rx}}(t) - P_{\text{rx}}(t-1)| \leq \Delta P_{\text{rx}}^{\text{max}}, \quad \forall t \geq 1, \quad (3)$$

$$\sum_{t \in \mathcal{T}} P_{\text{rx}}(t) \geq \overline{\text{CF}} \cdot \bar{P}_{\text{rx}} \cdot |\mathcal{T}|, \quad (4)$$

with $\bar{P}_{\text{rx}} = 870 \text{ MW}_{\text{th}}$, $\underline{P}_{\text{rx}} = 435 \text{ MW}_{\text{th}}$ (50% minimum-load fraction from the BWRX-300 design basis²¹), $\Delta P_{\text{rx}}^{\text{max}} = 522 \text{ MW}_{\text{th}} \text{ h}^{-1}$ corresponding to a 1 % min^{-1} ramp, and $\overline{\text{CF}} = 0.92$ from the mature U.S. light-water-reactor fleet average²². The annual capacity-factor constraint sets an annual energy floor equivalent to roughly one month of downtime, without modeling the timing of individual outages or introducing unit-commitment binaries.

The Rankine steam cycle is represented by a cascaded high-pressure to low-pressure turbine with a Willans-line extraction penalty. Letting $\eta_r = 0.345$ be the rated electric efficiency at zero extraction and $\alpha_w = 0.083 \text{ MW}_e \text{ MW}_{\text{th}}^{-1}$ the marginal electricity loss

intensity for mid-pressure steam diverted to the absorber:

$$P_{\text{tg}}(t) = \eta_r P_{\text{rx}}(t) - \alpha_w Q_{\text{ext}}(t), \quad (5)$$

$$P_{\text{tn}}(t) = (1 - \xi_{\text{aux}}) P_{\text{tg}}(t), \quad (6)$$

with $\xi_{\text{aux}} = 0.10$. Because diverted steam still expands through the high-pressure turbine before extraction, the Willans coefficient is roughly one order of magnitude smaller than charging the full turbine efficiency against extracted heat, as would be appropriate for a parallel main-steam tap.

Absorption-chiller submodel (Case 2). The double-effect LiBr-H₂O absorption chiller is described by a time-varying coefficient of performance and a binary availability indicator that captures LiBr crystallization risk^{18,19}. With $T_{\text{wb},0} = 26^\circ\text{C}$ as the Houston design wet-bulb, $\text{COP}_a^0 = 1.10$ as the COP at the design wet-bulb, $\delta = 0.015^\circ\text{C}^{-1}$ as the linear de-rating slope, and $\overline{\text{COP}}_a = 1.30$ as the nameplate cap, the chiller COP at wet-bulb $T_{\text{wb}}(t)$ is

$$\text{COP}_a(T_{\text{wb}}(t)) = \max \left\{ \min [\text{COP}_a^0 - \delta (T_{\text{wb}}(t) - T_{\text{wb},0}), \overline{\text{COP}}_a], 0.5 \right\}. \quad (7)$$

The lower bound of 0.5 is a numerical safeguard; Houston wet-bulbs in the weather record do not reduce a physical chiller to this value because the crystallization gate engages first. With $\Delta T_{\text{cw}} = 5\text{ K}$ as the cooling-tower approach, $T_{\text{cry}} = 32^\circ\text{C}$ as the LiBr-crystallization-line cooling-water inlet, and $a_m = 0.95$ a maintenance availability that derates the unit to account for scheduled servicing, the availability indicator is

$$a(t) = \begin{cases} a_m & \text{if } T_{\text{wb}}(t) + \Delta T_{\text{cw}} \leq T_{\text{cry}}, \\ 0 & \text{otherwise,} \end{cases} \quad (8)$$

so the absorption chiller is forced fully offline whenever the cooling-water inlet would exceed the LiBr crystallization line, and otherwise operates at its maintenance-derated availability. Equivalently, with the $\Delta T_{\text{cw}} = 5\text{ K}$ tower approach this is a wet-bulb gate at $T_{\text{cry}} - \Delta T_{\text{cw}} = 32 - 5 = 27^\circ\text{C}$, the value marked in Figure S1a. Delivered absorption cooling is

$$Q_a(t) = \text{COP}_a(T_{\text{wb}}(t)) \cdot a(t) \cdot Q_{\text{ext}}(t), \quad (9)$$

and a parasitic electric load $\beta_a Q_a(t)$ with $\beta_a = 0.020\text{ kW}_e\text{ kW}_c^{-1}$ powers solution and refrigerant pumps. The vapor-compression backup carries the residual cooling load whenever $a(t) = 0$.

Cooling, BESS and grid blocks. Vapor-compression cooling links its electric input to its cooling output through a part-load coefficient of performance. A water-cooled centrifugal chiller gains efficiency at reduced load, with the coefficient of performance peaking near 40–50% of rated duty before declining at very low load, following the integrated part-load value of ANSI/AHRI Standard 550/590²³. The chiller electricity is therefore a non-convex, piecewise-linear function of the cooling delivered,

450 $P_v(t) = f_v(Q_v(t))$, anchored at the Houston-derated full-load value $\text{COP}_v = 3.2$. In
 451 the closed-form Cases 0 and 3 the chiller is evaluated directly at the realized load
 452 fraction; in the nuclear Cases 1 and 2 the same curve enters the optimization as an
 453 SOS2 piecewise map, the single source of integer structure in the model. The lithium-
 454 ion BESS, sized at 100 MWh and 50 MW (C/2), follows the standard state-of-charge
 455 dynamics

$$E_B(t) = (1 - \rho_B) E_B(t-1) + (\sqrt{\eta_{rt}} B^+(t) - B^-(t)/\sqrt{\eta_{rt}}) \Delta t, \quad (10)$$

456 with initial state of charge fixed at $E_B(0) = 0.5 \text{ Cap}_E$ and cyclically restored through
 457 $E_B(|\mathcal{T}|) = E_B(0)$, $\eta_{rt} = 0.85$, $\rho_B = 4.17 \times 10^{-5} \text{ h}^{-1}$, and state-of-charge bounds
 458 $[0.05, 0.95] \text{ Cap}_E$. Grid exchange is bounded by the point of common coupling, $0 \leq$
 459 $P_g^+(t), P_g^-(t) \leq \bar{P}_{pcc}$, with $\bar{P}_{pcc} = 300 \text{ MW}$ for the baseline campus; in the size-
 460 matching sweep the interconnect is held at the baseline value for load multipliers
 461 below one and scales upward only for larger campuses, since downward-sizing the
 462 interconnect would artificially constrain exports from the fixed reactor.

463 **Energy balances.** Two equality constraints hold every hour. The electric balance
 464 equates supply, BESS discharge and grid imports to the data-center IT load, chiller
 465 drives, absorber parasitics, BESS charge and grid exports:

$$\begin{aligned} P_{tn}(t) + P_g^+(t) + B^-(t) &= P_{IT}(t) + P_v(t) + \beta_a Q_a(t) \\ &+ P_g^-(t) + B^+(t). \end{aligned} \quad (11)$$

466 The cooling balance equates total cooling supplied to the cooling demand of Eq. (1):

$$Q_a(t) + Q_v(t) = Q_{cool}^{dem}(t). \quad (12)$$

467 For Cases 0, 1 and 3 the absorption variables $Q_a(t)$, $Q_{ext}(t)$ are fixed to zero; for
 468 Case 3 the reactor and turbine variables are removed and the IT and chiller load is
 469 served entirely by the NGCC block at a full-load higher-heating-value (HHV) efficiency
 470 $\eta_{hhv} = 0.495$, with the part-load heat-rate penalty of Eq. (18).

471 **Objective.** The optimization minimizes TAC summed over seven accounting
 472 buckets:

$$\begin{aligned} \min \text{ TAC} &= \sum_k \text{CRF}_k \text{CAPEX}_k S_k + \text{FOM} + \text{VOM} \\ &+ \Phi_{\text{fuel}} + \Phi_{\text{grid}} + \Phi_{\text{CO}_2} - \Phi_{\text{ptc}}, \end{aligned} \quad (13)$$

473 where the first term annualizes overnight capital expenditure (CAPEX) across reac-
 474 tor, turbine, absorption and VCC chillers, BESS and NGCC sized at S_k , each
 475 amortized over its own engineering life N_k through an asset-specific capital recov-
 476 ery factor (CRF; reactor and turbine 20 yr, chillers 25 yr, NGCC 30 yr, battery
 477 15 yr); $\text{FOM} = \sum_k \text{FOM}_k S_k$ aggregates fixed operations and maintenance (FOM),
 478 and v_x denotes the variable operations and maintenance (VOM) coefficient of stream

479 x (Table S3); the variable terms are

$$\begin{aligned} \text{VOM} = & \sum_t [v_{\text{rx}} P_{\text{tn}}(t) + v_{\text{v}} Q_{\text{v}}(t) + v_{\text{a}} Q_{\text{a}}(t)] \Delta t \\ & + \sum_t v_{\text{B}} (B^+(t) + B^-(t)) \Delta t, \end{aligned} \quad (14)$$

$$\Phi_{\text{fuel}} = \pi_{\text{f}} \sum_t P_{\text{rx}}(t) \Delta t, \quad (15)$$

$$\Phi_{\text{grid}} = \sum_t \pi_{\text{g}}(t) [P_{\text{g}}^+(t) - P_{\text{g}}^-(t)] \Delta t, \quad (16)$$

$$\Phi_{\text{CO}_2} = \pi_{\text{CO}_2} \Theta \sum_t [\gamma_{\text{rx}} P_{\text{tn}}(t) + \theta(t) (P_{\text{g}}^+(t) - P_{\text{g}}^-(t))] \Delta t. \quad (17)$$

480 For Case 3, the same accounting buckets are used but the reactor terms are replaced
 481 by the NGCC terms. The combined-cycle heat rate rises below the design point, so
 482 fuel use and combustion emissions are both scaled by a part-load multiplier $\phi_{\text{ng}}(\ell) \geq$
 483 1 of the electrical load fraction $\ell(t) = P_{\text{ng}}(t)/\bar{P}_{\text{ng}}$, taken from a representative F-
 484 class combined-cycle characteristic and equal to one at full load. The VOM term uses
 485 $v_{\text{ng}} P_{\text{ng}}(t) + v_{\text{v}} Q_{\text{v}}(t)$, where $P_{\text{ng}}(t)$ is the NGCC electric output required by the IT and
 486 VCC loads, and the fuel and carbon terms become

$$\Phi_{\text{fuel}}^{\text{ng}} = \sum_t \pi_{\text{ng}}(t) h_{\text{ng}} \phi_{\text{ng}}(\ell(t)) P_{\text{ng}}(t) \Delta t, \quad (18)$$

$$\Phi_{\text{CO}_2}^{\text{ng}} = \pi_{\text{CO}_2} \Theta \sum_t \gamma_{\text{ng}} \phi_{\text{ng}}(\ell(t)) P_{\text{ng}}(t) \Delta t, \quad (19)$$

487 with full-load heat rate $h_{\text{ng}} = 3.412/\eta_{\text{hhv}}$ MMBtu MWh $^{-1}$ and full-load lifecycle factor
 488 $\gamma_{\text{ng}} = 420$ g CO $_2$ -eq kWh $^{-1}$, equal to 360 direct-combustion and 60 upstream-methane
 489 g CO $_2$ -eq kWh $^{-1}$ ²⁴. Applying the same ϕ_{ng} to both terms holds the fuel carbon con-
 490 tent constant. The carbon term Φ_{CO_2} multiplies the case carbon footprint by a uniform
 491 price π_{CO_2} and a unit-conversion factor $\Theta = 10^{-3}$ t CO $_2$ kg $^{-1}$. Exports credit the case
 492 at the ERCOT hourly average carbon intensity $\theta(t)$, derived from the U.S. Energy
 493 Information Administration (EIA)-930 generation mix²⁵, to reflect the time-varying
 494 emissions of the grid power exchanged, following time-resolved hourly emissions
 495 accounting rather than a static annual-average factor²⁶; this hourly average emission
 496 factor is used as a tractable proxy for the true marginal emission factor^{27,28}. Each asset
 497 is annualized over its own engineering life through $\text{CRF}_k = i(1+i)^{N_k}/[(1+i)^{N_k} - 1]$,
 498 with asset life N_k as listed above and baseline WACC $i = 6.7\%$; the battery addi-
 499 tionally carries a 5% end-of-life salvage credit. The WACC sensitivity in Section 2.4
 500 sweeps i over $\{5\%, 6.7\%, 10\%\}$.

501 **Nuclear production tax credit.** The nuclear cases earn the zero-emission nuclear
 502 production tax credit (Section 45U; 26 U.S.C. § 45U), enacted in the Inflation Reduc-
 503 tion Act of 2022²⁹, on their generation^{30,31}. At the prevailing-wage rate the credit is
 504 \$15 MWh $^{-1}$ (1.5¢ kWh $^{-1}$) and is subject to the statutory gross-receipts phaseout: it

holds at the full value while the hourly market price π_g stays at or below the threshold $\pi_1 = \$25 \text{ MWh}_e^{-1}$ (2.5¢ kWh_e⁻¹), then declines by \$0.80 for each \$1 MWh_e⁻¹ above it, reaching zero at $\pi_2 = \$43.75 \text{ MWh}_e^{-1}$ (4.375¢ kWh_e⁻¹),

$$c_{45U}(\pi) = \begin{cases} \bar{c}, & \pi \leq \pi_1, \\ \bar{c} \left(1 - \frac{\pi - \pi_1}{\pi_2 - \pi_1}\right), & \pi_1 < \pi < \pi_2, \\ 0, & \pi \geq \pi_2, \end{cases} \quad (20)$$

with full credit $\bar{c} = \$15 \text{ MWh}_e^{-1}$. The annual credit is applied to net nuclear generation,

$$\Phi_{\text{ptc}} = \sum_t c_{45U}(\pi_g(t)) P_{\text{tn}}(t) \Delta t, \quad (21)$$

using the hourly ERCOT price as the kWh-indexed gross-receipts proxy. The slope and breakpoints reproduce the statutory 16% reduction at the prevailing-wage multiplier. We treat the credit as a baseline policy condition, assuming a power-purchase-agreement structure in which the data center is an unrelated offtaker so that delivered nuclear generation qualifies; statutory amounts are inflation-indexed after 2024 and the program runs through tax year 2032, both of which we hold at their nominal in-window values. Cases 0 and 3 carry no nuclear generation and therefore no credit.

The cross-case figure of merit used throughout the rest of the paper is the grid-cost margin

$$M_c = 1 - \frac{\text{TAC}_c}{\text{TAC}_0} \Big|_{\text{matched scenario}}, \quad (22)$$

defined as one minus the ratio of the case TAC to the Case 0 (grid-only) TAC at the same year, PUE and carbon price. A positive margin means the configuration is cheaper than grid procurement and a negative margin means it is more expensive; the magnitude is the relative cost difference. We report it for all four configurations as a common measure of cost competitiveness against the grid.

4.3 Data inputs

Inputs to the optimization fall into three families: exogenous time series (hourly traces of IT load, electricity price, average carbon intensity and wet-bulb temperature), equipment parameters (capital and operating cost coefficients plus technical bounds for each plant block), and financial parameters (project lifetime and weighted-average cost of capital). All runs use 8760-hour annual operating horizons; leap-year raw files are trimmed to 8760 hourly records for cross-year comparability. The four characteristic input traces are shown in Supplementary Figure S1; aggregated values and source citations are listed in Supplementary Tables S2 and S3.

The Houston wet-bulb temperature distribution (Supplementary Figure S1a) drives the absorption-chiller availability constraint in Equation (8). Monthly box plots from the Open-Meteo historical-weather archive³² for the 2023 calendar year show that wet-bulb exceeds 26 °C (the absorption design anchor $T_{\text{wb},0}$) in about 32% of

summer hours and crosses the LiBr crystallization gate at 27 °C in about 5% of summer hours, concentrated in July and August afternoons. The ERCOT day-ahead LMP duration curves (Supplementary Figure S1b), retrieved through gridstatus.io³³, capture the recent range of grid-procurement cost. We label the years by annual-mean LMP and gas conditions rather than by single-hour price spikes: 2022 is the high annual-mean gas/high annual-mean LMP regime, 2023 the ATB-Mid baseline market year with moderate annual means, and 2024 the low annual-mean LMP regime. The IT-load profile (Supplementary Figure S1c) is built by scaling a 10 MW NLR colocation workload profile by 20× to the 200 MW facility scale and resampling it to hourly resolution³⁴; peak load reaches 142.4 MW_e and annual mean load is 94.7 MW_e. The absorption-chiller COP_a(T_{wb}) curve (Supplementary Figure S1d) parameterizes Equation (7): COP decreases linearly from the nameplate cap of 1.30 at cool ambient wet-bulb to 1.10 at the design anchor of 26 °C, after which the crystallization gate forces a binary cutoff and the VCC backup carries the cooling load.

Scalar inputs are summarized in Supplementary Table S2, and the per-asset cost and technical parameters carried through the CAPEX, FOM, and VOM terms in Equation (13) are listed in Supplementary Table S3. Reactor capital cost trajectories follow the NLR 2024 Annual Technology Baseline¹¹ at \$14,700, \$7,615 and \$2,250 kW_e⁻¹ for FOAK, ATB-Mid (NLR Moderate) and NOAK cases. The absorption-chiller capital cost grid is anchored at \$750 kW_c⁻¹ baseline and brackets the literature range from \$450 kW_c⁻¹ (Bare-Low, Chinese manufacturer ex-works) to \$1,200 kW_c⁻¹ (Turnkey-High, Western turnkey engineering, procurement and construction). The NGCC cost block follows the NLR ATB 2024 F-class ATB-Mid (NLR Moderate) values; natural-gas fuel cost is the U.S. Energy Information Administration daily Henry Hub spot price³⁵, forward-filled where needed and applied to each hourly dispatch interval, plus the Houston Ship Channel basis (mean \$0.10 MMBtu⁻¹). The data audit trail and provenance summary are provided in the Supplementary Information.

4.4 Scenario design

Eight sensitivity groups are evaluated relative to the baseline 2023, full-load PUE 1.35, ATB-Mid CAPEX and WACC 6.7% configuration to probe the boundary conditions of the cogeneration economics. The full run ledger, consisting of four baseline case runs plus 96 sensitivity runs, is given in Supplementary Table S9.

The cooling-efficiency sweep tests the sensitivity of the absorption value proposition to the electric cooling efficiency, implemented by changing the VCC full-load COP and reporting the equivalent full-load PUE while holding the physical heat-rejection load fixed at Eq. (1). The year sweep covers the recent range of Henry Hub and ERCOT prices. The BESS sweep tests whether 100 MWh of storage adds value at current Li-ion capital costs in the nuclear cases, with Case 0 and Case 3 retained as no-storage control rows. The SMR CAPEX sweep covers the published FOAK-to-NOAK range. The two-dimensional SMR × absorption CAPEX grid (group G5) is the central scenario: it asks whether any joint capital-cost trajectory makes Case 2 cheaper than Case 0 at the 2023 baseline. The carbon-price sweep identifies the carbon price at which the nuclear cases cross the grid-only baseline. The WACC sweep isolates the financing effect on Case 2

580 with the operating parameters held constant. The data-center size sweep (group G8)
581 scales the IT-load trace, data-center-side cooling and NGCC capacities from $0.5\times$ to
582 $3.0\times$ while holding the commercial BWRX-300 unit fixed; the PCC remains at the
583 300 MW baseline for load multipliers below one and scales upward only for larger
584 campuses, isolating unit-size matching without artificially constraining exports from
585 the fixed reactor.

586 Data availability

587 All data that support the findings of this study, including the input parameters, sce-
588 nario definitions, run-level summaries, hourly dispatch traces and figure source data,
589 are available in the project GitHub repository at <https://github.com/Henryutd2021/Nuclear-DC> and will be deposited in a persistent public archive (Zenodo) with a
590 citable DOI on acceptance. The individual exogenous data sources and their prove-
591 nance are listed in Supplementary Note 4. No proprietary or restricted-access data are
592 used.
593

594 Code availability

595 The optimization framework, the scenario driver and the figure-generation notebooks
596 that reproduce every result reported here are available in the same GitHub repository
597 and will be archived alongside the data on acceptance.

598 Acknowledgements

599 Author contributions

600 H.L. conceived the study, developed the optimization framework and the hourly
601 absorption submodel, curated the input data, ran the scenarios, produced the figures
602 and wrote the original draft. B.S. contributed to the methodology, validated the
603 optimization results, and reviewed and edited the manuscript. J.Z. conceived and
604 supervised the study, provided resources and funding, and reviewed and edited the
605 manuscript. All authors approved the final manuscript.

606 Competing interests

607 The authors declare no competing interests.

608 References

- 609 [1] Masanet E, Shehabi A, Lei N, Smith S, Koomey J. Recalibrating global data
610 center energy-use estimates. *Science*. 2020;367(6481):984–986. <https://doi.org/10.1126/science.aba3758>.
611
- 612 [2] Shehabi A, et al. 2024 United States Data Center Energy Usage Report. Berkeley,
613 CA: Lawrence Berkeley National Laboratory; 2024. Report to Congress under the

- 614 Energy Act of 2020. Available from: https://eta-publications.lbl.gov/sites/default/files/2024-12/lbnl-2024-united-states-data-center-energy-usage-report_1.pdf.
615
- 616 [3] International Energy Agency. Electricity 2024: Analysis and forecast to 2026.
617 Paris: International Energy Agency; 2024. Available from: <https://www.iea.org/reports/electricity-2024>.
618
- 619 [4] International Energy Agency. Energy and AI. Paris: International Energy
620 Agency; 2025. Available from: <https://www.iea.org/reports/energy-and-ai>.
- 621 [5] Electric Reliability Council of Texas. Long-Term Load Forecast Update (2025–
622 2031) and Methodology Changes. Austin, TX: ERCOT; 2025. Board of Directors
623 meeting materials, Item 8.1, April 2025. Available from: <https://www.ercot.com/files/docs/2025/04/07/8.1-Long-Term-Load-Forecast-Update-2025-2031-and-Methodology-Changes.pdf>.
624
625
- 626 [6] U S Energy Information Administration. Annual Energy Outlook 2026. Wash-
627 ington, DC: U.S. Energy Information Administration; 2026. Available from:
628 <https://www.eia.gov/outlooks/aeo/>.
- 629 [7] The Green Grid. PUE: A Comprehensive Examination of the Metric. The Green
630 Grid; 2012. White Paper 49. Available from: <https://www.thegreengrid.org/en/resources/library-and-tools/237-WP%2349---PUE%3A-A-Comprehensive-Examination-of-the-Metric>.
631
632
- 633 [8] ASHRAE Technical Committee 9.9. Thermal Guidelines for Data Processing
634 Environments. 5th ed. Peachtree Corners, GA: ASHRAE; 2021.
- 635 [9] Mignacca B, Locatelli G. Economics and finance of Small Modular Reactors:
636 A systematic review and research agenda. Renewable and Sustainable Energy
637 Reviews. 2020;118:109519. <https://doi.org/10.1016/j.rser.2019.109519>.
- 638 [10] U S Department of Energy. Pathways to Commercial Liftoff: Advanced Nuclear.
639 Washington, DC: U.S. Department of Energy; 2024. September 2024 update.
640 Available from: <https://liftoff.energy.gov/advanced-nuclear-2/>.
- 641 [11] National Laboratory of the Rockies. 2024 Annual Technology Baseline: Nuclear.
642 Golden, CO: National Laboratory of the Rockies; 2024. Available from: <https://atb.nlr.gov/electricity/2024/nuclear>.
643
- 644 [12] Asuega A, Limb BJ, Quinn JC. Techno-economic analysis of advanced small
645 modular nuclear reactors. Applied Energy. 2023;334:120669. <https://doi.org/10.1016/j.apenergy.2023.120669>.
646
- 647 [13] Idaho National Laboratory. Meta-Analysis of Advanced Nuclear Reactor Cost
648 Estimations. Idaho Falls, ID: Idaho National Laboratory; 2024. INL/RPT-24-
649 77048, Rev. 0. Available from: <https://gain.inl.gov/content/uploads/4/2024/11>

650 /INL-RPT-24-77048-Meta-Analysis-of-Adv-Nuclear-Reactor-Cost-Estimations
651 .pdf.

652 [14] OECD Nuclear Energy Agency. Unlocking Reductions in the Construction Costs
653 of Nuclear: A Practical Guide for Stakeholders. Paris: OECD Publishing; 2020.
654 Available from: [https://www.oecd.org/en/publications/unlocking-reductions-i](https://www.oecd.org/en/publications/unlocking-reductions-in-the-construction-costs-of-nuclear_33ba86e1-en.html)
655 [n-the-construction-costs-of-nuclear_33ba86e1-en.html](https://www.oecd.org/en/publications/unlocking-reductions-in-the-construction-costs-of-nuclear_33ba86e1-en.html).

656 [15] Ebrahimi K, Jones GF, Fleischer AS. A review of data center cooling technol-
657 ogy, operating conditions and the corresponding low-grade waste heat recovery
658 opportunities. Renewable and Sustainable Energy Reviews. 2014;31:622–638.
659 <https://doi.org/10.1016/j.rser.2013.12.007>.

660 [16] Yuan X, Liang Y, Hu X, Xu Y, Chen Y, Kosonen R. Waste heat recoveries in data
661 centers: A review. Renewable and Sustainable Energy Reviews. 2023;188:113777.
662 <https://doi.org/10.1016/j.rser.2023.113777>.

663 [17] Amiri L, Madadian E, Bahrani N, Ghoreishi-Madiseh SA. Techno-Economic
664 Analysis of Waste Heat Utilization in Data Centers: Application of Absorption
665 Chiller Systems. Energies. 2021;14(9):2433. <https://doi.org/10.3390/en14092433>.

666 [18] Srihirin P, Aphornratana S, Chungpaibulpatana S. A review of absorp-
667 tion refrigeration technologies. Renewable and Sustainable Energy Reviews.
668 2001;5(4):343–372. [https://doi.org/10.1016/S1364-0321\(01\)00003-X](https://doi.org/10.1016/S1364-0321(01)00003-X).

669 [19] Herold KE, Radermacher R, Klein SA. Absorption Chillers and Heat Pumps.
670 2nd ed. Boca Raton, FL: CRC Press; 2016.

671 [20] Cui Z, Du S, Wang R. Switchable multistage absorption chiller for ultra-low-
672 grade waste heat driven cooling in data centers. Applied Energy. 2025;398:126407.
673 <https://doi.org/10.1016/j.apenergy.2025.126407>.

674 [21] GE Vernova Hitachi Nuclear Energy.: BWRX-300 Small Modular Reactor. Prod-
675 uct information page. Available from: [https://www.gevernova.com/nuclear/ca](https://www.gevernova.com/nuclear/carbon-free-power/bwr-x-300-small-modular-reactor)
676 [rbon-free-power/bwr-x-300-small-modular-reactor](https://www.gevernova.com/nuclear/carbon-free-power/bwr-x-300-small-modular-reactor).

677 [22] U S Energy Information Administration.: Electric Power Monthly, Table 6.07.B:
678 Capacity Factors for Utility Scale Generators Primarily Using Non-Fossil Fuels.
679 U.S. Energy Information Administration, Washington, DC. Available from: [https:](https://www.eia.gov/electricity/monthly/epm_table_grapher.php?t=epmt_6_07_b)
680 [//www.eia.gov/electricity/monthly/epm_table_grapher.php?t=epmt_6_07_b](https://www.eia.gov/electricity/monthly/epm_table_grapher.php?t=epmt_6_07_b).

681 [23] Air-Conditioning, Heating, and Refrigeration Institute. ANSI/AHRI Standard
682 550/590 (I-P): Performance Rating of Water-Chilling and Heat Pump Water-
683 Heating Packages Using the Vapor Compression Cycle. Arlington, VA: AHRI;
684 2020.

- [24] Alvarez RA, et al. Assessment of methane emissions from the U.S. oil and gas supply chain. *Science*. 2018;361(6398):186–188. <https://doi.org/10.1126/science.aar7204>.
- [25] U S Energy Information Administration.: Hourly Electric Grid Monitor (Form EIA-930). U.S. Energy Information Administration, Washington, DC. Available from: <https://www.eia.gov/electricity/gridmonitor/>.
- [26] Li H, Shen G, Senemmar S, Mehmani A, Zhang J. Real-time greenhouse gas emission intensity informed demand-side load regulation for power grid decarbonization. *Cell Reports Sustainability*. 2025;2(5):100367. <https://doi.org/10.1016/j.crsus.2025.100367>.
- [27] Hawkes AD. Estimating marginal CO₂ emissions rates for national electricity systems. *Energy Policy*. 2010;38(10):5977–5987. <https://doi.org/10.1016/j.enpol.2010.05.053>.
- [28] Siler-Evans K, Azevedo IL, Morgan MG, Apt J. Regional variations in the health, environmental, and climate benefits of wind and solar generation. *Proceedings of the National Academy of Sciences*. 2013;110(29):11768–11773. <https://doi.org/10.1073/pnas.1221978110>.
- [29] United States Congress.: Inflation Reduction Act of 2022. Public Law No. 117–169, 136 Stat. 1818. Available from: <https://www.congress.gov/bill/117th-congress/house-bill/5376/text>.
- [30] U S Code.: 26 U.S.C. § 45U: Zero-emission nuclear power production credit. Office of the Law Revision Counsel; Inflation Reduction Act of 2022, Pub. L. 117–169, § 13105. Available from: [https://uscode.house.gov/view.xhtml?req=\(title:26+section:45U+edition:prelim\)](https://uscode.house.gov/view.xhtml?req=(title:26+section:45U+edition:prelim)).
- [31] Internal Revenue Service.: Zero-Emission Nuclear Power Production Credit (Section 45U), Form 7213. U.S. Department of the Treasury. Available from: <https://www.irs.gov/credits-deductions/zero-emission-nuclear-power-production-credit>.
- [32] Zippenfenig P.: Open-Meteo.com Weather API. Zenodo. Available from: <https://open-meteo.com/>.
- [33] gridstatus io.: gridstatus: Open Electricity Grid and Market Data. Open-source Python library and data platform. Available from: <https://www.gridstatus.io/>.
- [34] Vercellino R, Willard J, Campos G, da Silva Pereira W, Hull O, Selensky M, et al. Measurement of Generative AI Workload Power Profiles for Whole-Facility Data Center Infrastructure Planning. Golden, CO: National Laboratory of the Rockies Data Catalog; 2026. DOI: 10.7799/3025227; arXiv:2604.07345. Dataset and technical report. Available from: <https://arxiv.org/abs/2604.07345>.

- 722 [35] U S Energy Information Administration.: Henry Hub Natural Gas Spot Price.
723 U.S. Energy Information Administration, Washington, DC. Available from: [https:](https://www.eia.gov/dnav/ng/hist/rngwhhdD.htm)
724 [//www.eia.gov/dnav/ng/hist/rngwhhdD.htm](https://www.eia.gov/dnav/ng/hist/rngwhhdD.htm).
- 725 [36] United Nations Economic Commission for Europe. Carbon Neutrality in the
726 UNECE Region: Integrated Life-cycle Assessment of Electricity Sources. Geneva:
727 United Nations Economic Commission for Europe; 2022. Available from: [https:](https://unece.org/sed/documents/2021/10/reports/life-cycle-assessment-electricity-generation-options)
728 [//unece.org/sed/documents/2021/10/reports/life-cycle-assessment-electricity-g](https://unece.org/sed/documents/2021/10/reports/life-cycle-assessment-electricity-generation-options)
729 [eneration-options](https://unece.org/sed/documents/2021/10/reports/life-cycle-assessment-electricity-generation-options).

Supplementary Information

Supplementary Note 1: Nomenclature

The complete set of model sets, decision variables, parameters and abbreviations is listed in Supplementary Table S1.

Table S1: Nomenclature for the optimization model: sets and indices, decision variables, parameters and abbreviations.

Symbol	Description
Sets and indices	
$t \in \mathcal{T}$	hourly time step over an annual modeling horizon, $ \mathcal{T} = 8760$
$c \in \mathcal{C}$	system configuration, $\mathcal{C} = \{0, 1, 2, 3\}$
Decision variables (hour t)	
$P_{\text{rx}}(t)$	reactor thermal output (MW_{th})
$P_{\text{tg}}(t)$	turbine gross electric output (MW_{e})
$P_{\text{tn}}(t)$	turbine net electric output (MW_{e})
$Q_{\text{ext}}(t)$	mid-pressure steam extraction to absorber (MW_{th})
$Q_{\text{a}}(t)$	absorption-chiller cooling delivered (MW_{c})
$P_{\text{v}}(t)$	vapor-compression chiller (VCC) electric input (MW_{e})
$Q_{\text{v}}(t)$	VCC cooling delivered (MW_{c})
$P_{\text{g}}^{+}(t)$	grid import (MW_{e})
$P_{\text{g}}^{-}(t)$	grid export (MW_{e})
$P_{\text{ng}}(t)$	NGCC electric output, Case 3 (MW_{e})
$B^{+}(t)$	BESS charging power (MW_{e})
$B^{-}(t)$	BESS discharging power (MW_{e})
$E_{\text{B}}(t)$	BESS state of charge (MWh)
Parameters	
$P_{\text{IT}}(t)$	data-center IT load (MW_{e})
$\pi_{\text{g}}(t)$	ERCOT day-ahead LMP ($\$ \text{MWh}_{\text{e}}^{-1}$)
$\theta(t)$	ERCOT hourly average carbon intensity ($\text{g CO}_2 \text{ kWh}_{\text{e}}^{-1}$)
$T_{\text{wb}}(t)$	Houston wet-bulb temperature ($^{\circ}\text{C}$)
η_{r}	turbine rated electric efficiency at zero extraction ($-$)
α_{w}	Willans extraction penalty ($\text{MW}_{\text{e}} \text{ MW}_{\text{th}}^{-1}$)
ξ_{aux}	turbine auxiliary fraction ($-$)
PUE	power usage effectiveness ($-$)

continued on next page

Supplementary Table S1 (continued)

Symbol	Description
η_{ch}	chilled-water chain efficiency (–)
$\text{COP}_{\text{a}}(T_{\text{wb}})$	absorption chiller COP (–)
$a(t)$	absorption availability (a_{m} operating, 0 crystallization)
a_{m}	absorption maintenance availability (–)
COP_{v}	VCC full-load coefficient of performance (–)
$f_{\text{v}}(\cdot)$	VCC part-load electricity map (MW_{e})
β_{a}	absorber parasitic load intensity ($\text{kW}_{\text{e}} \text{ kW}_{\text{c}}^{-1}$)
η_{rt}	BESS round-trip efficiency (–)
ρ_{B}	BESS hourly self-discharge (–)
Cap_{E}	BESS energy capacity (MWh)
$\overline{\text{CF}}$	reactor annual capacity factor (–)
$\overline{P}_{\text{rx}}, \underline{P}_{\text{rx}}$	reactor maximum and minimum thermal output (MW_{th})
$\Delta P_{\text{rx}}^{\text{max}}$	reactor ramp limit ($\text{MW}_{\text{th}} \text{ h}^{-1}$)
$\overline{P}_{\text{pcc}}$	point-of-common-coupling capacity (MW_{e})
CAPEX_k	overnight capital coefficient of asset k
FOM_k	fixed O&M coefficient of asset k
VOM_k	variable O&M coefficient of asset k
CRF_k	capital recovery factor of asset k (–)
N_k	engineering life of asset k (yr)
π_{f}	delivered nuclear fuel cost ($\$ \text{ MWh}_{\text{th}}^{-1}$)
$\pi_{\text{ng}}(t)$	delivered natural-gas price for NGCC (Henry Hub + basis; $\$ \text{ MMBtu}^{-1}$)
h_{ng}	NGCC full-load HHV heat rate ($\text{MMBtu} \text{ MWh}_{\text{e}}^{-1}$)
η_{hhv}	NGCC full-load HHV net efficiency (–)
$\overline{P}_{\text{ng}}$	NGCC rated electric capacity (MW_{e})
$\phi_{\text{ng}}(\ell)$	NGCC part-load heat-rate multiplier (–)
$\bar{c}, \pi_1, \pi_2$	45U full credit and phaseout breakpoints ($\$ \text{ MWh}_{\text{e}}^{-1}$)
π_{CO_2}	carbon price ($\$ \text{ tCO}_2^{-1}$)
Δt	hourly time step length (h)
γ_{rx}	reactor lifecycle emissions ($\text{g CO}_2 \text{ kWh}_{\text{e}}^{-1}$)
γ_{ng}	NGCC lifecycle emissions ($\text{g CO}_2\text{-eq} \text{ kWh}_{\text{e}}^{-1}$)
Θ	kg-to-tonne unit conversion ($= 10^{-3}$)
Abbreviations	
ATB	National Laboratory of the Rockies Annual Technology Baseline

continued on next page

Supplementary Table S1 (continued)

Symbol	Description
BWRX-300	GE–Hitachi 870 MW _{th} , 270 MW _e Boiling Water small modular reactor
CAPEX	overnight capital expenditure
CRF	capital recovery factor
COP	coefficient of performance
ERCOT	Electric Reliability Council of Texas
FOAK–NOAK	first- and nth-of-a-kind reactor capital cost
FOM, VOM	fixed and variable operations and maintenance
IT	information-technology load
M	grid-cost margin, $1 - \text{TAC}_c / \text{TAC}_0$ (positive = cheaper than grid)
LiBr–H ₂ O	lithium-bromide–water absorption pair
LMP	locational marginal price
NGCC	natural gas combined cycle
PTC	production tax credit (Inflation Reduction Act Section 45U)
PUE	power usage effectiveness
SMR	small modular reactor
TAC	total annualized cost (\$ yr ⁻¹)
VCC	vapor-compression chiller
WACC	weighted-average cost of capital

Supplementary Note 2: Input data and equipment parameters

The scalar input data and the per-asset cost and performance parameters carried through the optimization are listed in Supplementary Tables S2 and S3.

Supplementary Note 3: Input traces

The four characteristic exogenous input traces driving the optimization are shown in Supplementary Figure S1.

Supplementary Note 4: Exogenous data sources and provenance

Each exogenous input is drawn from a public, citable source and trimmed to an 8760-hour annual horizon. ERCOT day-ahead locational marginal prices at the Houston load zone are retrieved through the gridstatus.io public data interface³³. Hourly average grid carbon intensities are derived from the U.S. Energy Information Administration EIA-930 Hourly Electric Grid Monitor generation mix²⁵ weighted by United Nations Economic Commission for Europe 2022 lifecycle emission factors for each technology³⁶. Houston wet-bulb temperature is taken from the Open-Meteo historical-weather reanalysis archive³². The IT-load trace is derived from the Vercellino et al.

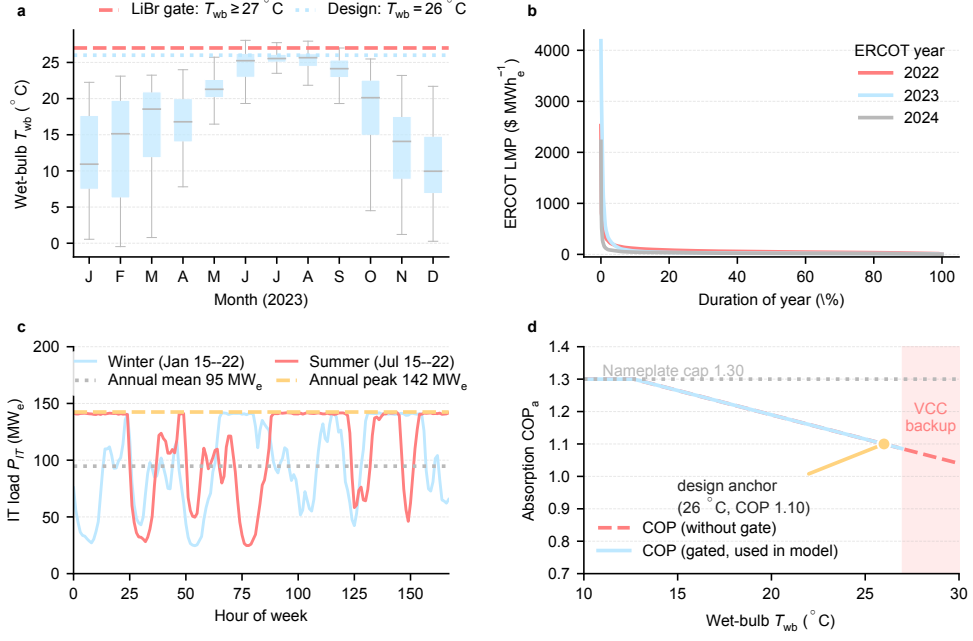


Fig. S1 Hourly input traces driving the optimization. LiBr denotes lithium bromide; COP, coefficient of performance; ERCOT, Electric Reliability Council of Texas; LMP, locational marginal price; IT, information technology; VCC, vapor-compression chiller. (a) Houston monthly wet-bulb distribution from the Open-Meteo historical-weather archive (year 2023). The dashed horizontal line marks the LiBr crystallization gate at $T_{wb} \geq 27^{\circ}\text{C}$; the dotted line marks the COP design anchor $T_{wb,0} = 26^{\circ}\text{C}$. (b) Duration curves of ERCOT day-ahead LMP for the three modeled years; the regime labels refer to annual-mean LMP and gas conditions, not to single-hour price spikes. (c) Data-center IT-load profile for a typical winter and summer week, with annual mean and peak overlaid. (d) Double-effect LiBr- H_2O absorption-chiller COP as a function of wet-bulb temperature; the dashed segment shows the linear de-rating below the crystallization gate, and the shaded band marks the backup-cooling region in which the VCC carries the cooling load.

2026 NLR Data Catalog colocation profile, scaling the 10 MW source profile by $20\times$ to represent a 200 MW facility³⁴. Delivered natural gas is the U.S. Energy Information Administration Henry Hub daily spot price³⁵ plus a fixed Houston Ship Channel basis. Reactor and NGCC overnight capital costs and fixed operations and maintenance follow the National Laboratory of the Rockies 2024 Annual Technology Baseline¹¹; the reactor capacity factor is the U.S. Energy Information Administration fleet average²²; and absorption-chiller capital costs are bracketed from American Society of Heating, Refrigerating and Air-Conditioning Engineers Technical Committee 8.3 absorption-chiller technical data and publicly available vendor literature. The Section 45U production tax credit follows the enacting statute^{29,30} and U.S. Internal Revenue Service guidance³¹. No proprietary or restricted-access data are used, and the audit trail linking each raw file to its processed model input is included in the project repository.

Supplementary Note 5: Market inputs by operating year

The three modeled years bracket the recent range of ERCOT and natural-gas market conditions (Supplementary Table S4). The table is included to document why 2023 is the baseline operating year rather than to repeat the input-trace panel: 2022 has high annual-mean gas and high annual-mean LMP, 2023 has moderate annual means with low gas but firm power prices, and 2024 has low annual-mean LMP that makes grid procurement comparatively cheap. The labels are annual-mean market descriptors and do not imply that 2023 lacked short-duration price spikes. The on-site natural gas combined cycle (Case 3) beats grid supply only in 2023, when low delivered gas coincides with firm ERCOT prices; in 2022 the gas and grid prices move together and erase the advantage, and in 2024 the low ERCOT price makes grid procurement the cheapest option.

Supplementary Note 6: Cogeneration dispatch and storage detail

In the 2023 ATB-Mid baseline, the absorption chiller delivers 862 GWh_c of cooling and the vapor-compression backup 60 GWh_c. The backup energy is spread over roughly 2,870 hours of partial top-up, which also cover the 5% maintenance derating of the absorber. The crystallization gate of Equation (8) forces the absorber fully offline for about 115 hours, concentrated from late June through August, during which the vapor-compression chiller carries the entire 13.6 GWh_c cooling load of those hours. In the islanded gas case (Case 3) the 2023 day-ahead price falls below the NGCC running cost of about \$21 MWh_e⁻¹ in roughly 42% of hours, but the disconnected plant cannot import in those hours, so its comparison against grid supply is made on annual cost rather than hour by hour. The lithium-ion battery sweep (group G3) lowers total annualized cost by \$0.63 M yr⁻¹ in each nuclear case, an improvement of about 0.8 percentage points in the grid-cost margin that never changes the configuration ranking, so it is reported as a secondary boundary rather than a main result.

Supplementary Note 7: Marginal value of absorption cooling

Supplementary Figure S2 and Supplementary Table S7 report the marginal value of adding the absorption chiller to the SMR-only plant (Case 2 minus Case 1) at the 2023 ATB-Mid baseline and across the cooling-efficiency sweep. The figure shows the trend and the related margin response, while the table provides the exact component values. The gross avoided vapor-compression electricity grows with the reported PUE, so the chiller moves from a net penalty against an efficient electric chiller (PUE 1.10) to a net saving against an inefficient one (PUE 1.50), with break-even near a full-load PUE of 1.40. In the high annual-mean gas/high annual-mean LMP 2022 conditions, the standalone net value of absorption is approximately $-\$0.03 \text{ M yr}^{-1}$, showing that heat recovery is market-regime dependent rather than uniformly positive.

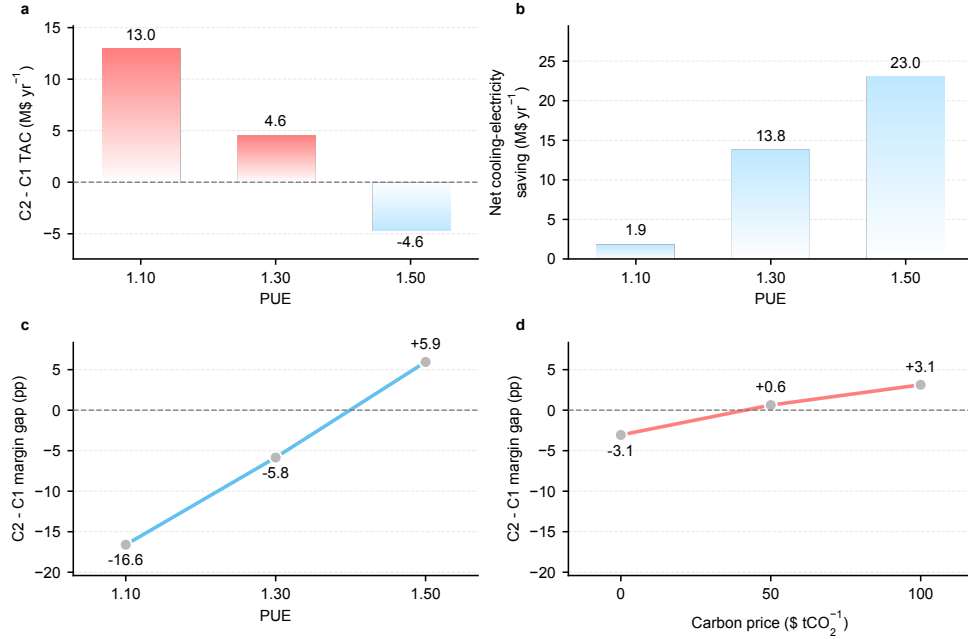


Fig. S2 Cooling-side response of the absorption configuration at the 2023 ATB-Mid baseline. ATB-Mid denotes the mid-range National Laboratory of the Rockies Annual Technology Baseline reactor-capital case; TAC, total annualized cost; PUE, power usage effectiveness; VCC, vapor-compression chiller. (a) Case 2 minus Case 1 TAC difference across full-load PUE; positive is a cogeneration penalty, negative a saving. (b) Net cooling-electricity saving from absorption, measured as gross avoided VCC electricity cost less Case 2 VCC backup/top-up, before fixed absorption capital and turbine-opportunity costs. (c) The same difference expressed as a grid-cost-margin gap across full-load PUE. (d) The margin gap across carbon price. Absorption becomes economic only when the competing electric chiller is inefficient (high PUE) or carbon is priced.

Supplementary Note 8: Data-center size-matching results

Supplementary Figure S3 and Supplementary Table S8 report the size-matching sweep (group G8). Holding the BWRX-300 reactor fixed at 270 MW_e net while scaling the IT load, cooling and NGCC capacities from 0.5 $\times$ to 3.0 $\times$ moves Case 2 from a deep merchant-export surplus at small scale to a net importer at 3.0 $\times$; the PCC is held at the 300 MW baseline for sub-baseline load multipliers and scales upward only for larger campuses. The annual dollar gap to grid supply is nearly flat because the larger campus also raises the grid-only counterfactual; what improves is the IT-energy-normalized cost intensity, which falls from $\$235$ to $\$143 \text{ MWh}_{\text{IT}}^{-1}$.

Supplementary Note 9: Scenario run ledger

The baseline and eight sensitivity groups, comprising the four baseline case runs and the 96 sensitivity runs evaluated in this study, are listed in Supplementary Table S9.

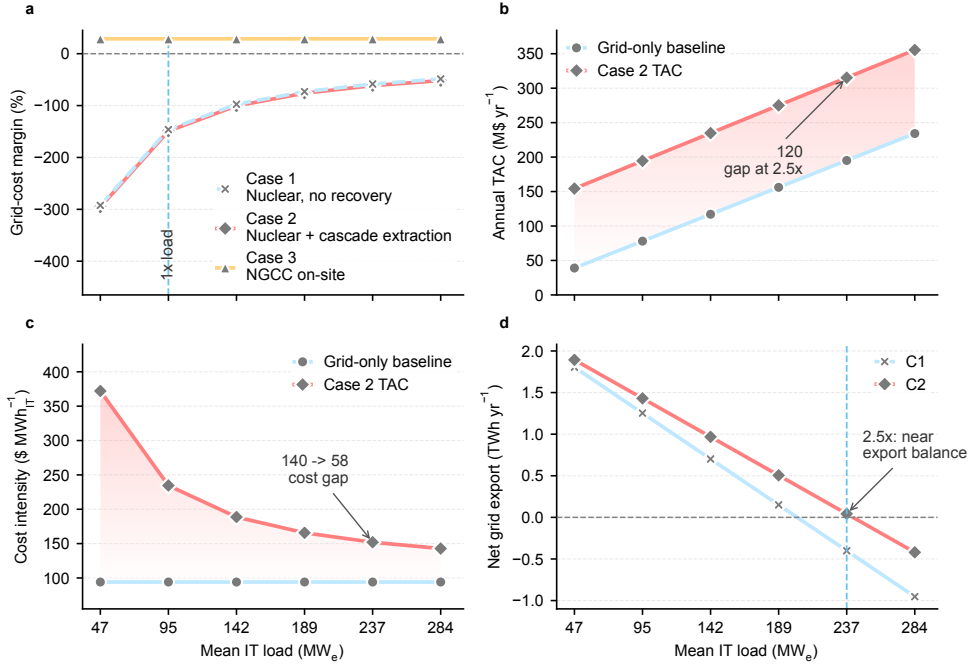


Fig. S3 Data-center size matching sensitivity (G8). IT denotes information technology; NGCC, natural gas combined cycle; TAC, total annualized cost. The IT-load trace and data-center-side cooling and NGCC capacities are scaled from $0.5\times$ to $3.0\times$ while the BWRX-300 reactor remains fixed at 270 MW_e net; the PCC is held at 300 MW below $1\times$ and scales upward above $1\times$. (a) TAC-normalized grid-cost margin for the on-site cases against the matching grid-only baseline. (b) Case 2 TAC against the matching grid-only TAC, with the shaded area indicating the annual dollar gap. (c) IT-energy-normalized cost intensity; the shaded area is the excess Case 2 cost over grid supply. (d) Net grid export for the two nuclear cases; positive values indicate merchant export and negative values indicate net grid import. Arrow labels highlight the $2.5\times$ load reference discussed in the text.

Table S2 Input data sources, baseline values and units. ERCOT denotes Electric Reliability Council of Texas; LMP, locational marginal price; IT, information technology; NLR ATB, National Laboratory of the Rockies Annual Technology Baseline; FOAK, first-of-a-kind; NOAK, nth-of-a-kind; O&M, operations and maintenance; VCC, vapor-compression chiller; BESS, battery energy storage system; NGCC, natural gas combined cycle; WACC, weighted-average cost of capital; PTC, production tax credit.

Quantity	Baseline value or range	Resolution	Source
ERCOT day-ahead LMP, Houston Hub	yearly mean \$28–70 MWh_e^{-1} (2022–2024)	hourly	gridstatus.io
ERCOT average carbon intensity, ERCOT system	320–351 $\text{g CO}_2 \text{ kWh}_e^{-1}$ annual mean (hourly average emission factor)	hourly	EIA-930 + United Nations Economic Commission for Europe 2022
Delivered natural gas (Henry Hub + basis)	\$2.35–6.53 MMBtu^{-1} (annual mean)	daily, applied hourly	U.S. Energy Information Administration Henry Hub spot
Houston wet-bulb temperature	−0.5 to 28 $^{\circ}\text{C}$ annual range	hourly	Open-Meteo historical
Data-center IT load	142.4 MW_e peak; 94.7 MW_e mean	hourly	Vercellino et al. 2026 / NLR Data Catalog, 10 MW colocation profile scaled $20\times$ ³⁴
BWRX-300 capital cost	\$7,615 kW_e^{-1} ATB-Mid (NLR Moderate); \$14,700 kW_e^{-1} FOAK; \$2,250 kW_e^{-1} NOAK	scalar	NLR ATB 2024 ¹¹
BWRX-300 fixed O&M	\$121 $\text{kW}_e^{-1} \text{ yr}^{-1}$	scalar	NLR ATB 2024
BWRX-300 capacity factor	0.92 (fleet average)	scalar	U.S. Energy Information Administration Electric Power Monthly Table 6.07.B ²²
BWRX-300 lifecycle emissions	12 $\text{g CO}_2\text{-eq kWh}_e^{-1}$	scalar	United Nations Economic Commission for Europe 2022
Absorption chiller capital cost	\$450–1,200 kW_c^{-1} (\$750 baseline)	scalar	American Society of Heating, Refrigerating and Air-Conditioning Engineers Technical Committee 8.3 + vendor literature
VCC capital cost	\$250 kW_c^{-1}	scalar	American Society of Heating, Refrigerating and Air-Conditioning Engineers typical centrifugal
Li-ion BESS capital cost	\$405 kWh^{-1} + \$1,620 kW^{-1}	scalar	NLR ATB 2024
NGCC capital cost	\$1,330 kW_e^{-1}	scalar	NLR ATB 2024 F-class
NGCC direct CO_2	360 $\text{g CO}_2 \text{ kWh}_e^{-1}$ + 60 g upstream CH_4 ($\text{CO}_2\text{-eq}$)	scalar	ATB-Mid (NLR Moderate)
WACC	6.7% baseline; {5, 6.7, 10}% in scenario group G7	scalar	U.S. Energy Information Administration + Alvarez et al. (2018)
Asset amortization life	20 yr reactor and turbine; 25 yr chillers; 30 yr NGCC; 15 yr battery	scalar	Massachusetts Institute of Technology nuclear-finance survey
Nuclear PTC (Section 45U)	\$15 MWh_e^{-1} full credit; phaseout \$25–43.75 MWh_e^{-1}	scalar	NLR ATB 2024
			26 U.S.C. 45U; U.S. Internal Revenue Service Form 7213

Table S3 Detailed equipment cost and performance parameters used in the optimization. CAPEX denotes overnight capital expenditure; FOM, fixed operations and maintenance; VOM, variable operations and maintenance; VCC, vapor-compression chiller; BESS, battery energy storage system; NGCC, natural gas combined cycle; HHV, higher heating value; ERCOT, Electric Reliability Council of Texas; PCC, point of common coupling. Each asset’s CAPEX is annualized over its own listed lifetime through an asset-specific capital recovery factor; the battery additionally carries a 5% end-of-life salvage credit.

Cost and lifetime inputs.

Asset	CAPEX	FOM	VOM	Life
BWRX-300 reactor (Cases 1, 2)	\$7,615 kW _e ⁻¹	\$121 kW _e ⁻¹ yr ⁻¹	\$3.50 MWh _e ⁻¹	20 yr
Cascaded steam turbine (Cases 1, 2)	bundled into reactor overnight capital cost	bundled	bundled	20 yr
Double-effect absorption chiller (Case 2)	\$450–1,200 kW _c ⁻¹ (\$750 baseline)	\$20 kW _c ⁻¹ yr ⁻¹	\$0.40 MWh _c ⁻¹	25 yr
Vapor-compression chiller (Cases 0, 1, 2, 3)	\$250 kW _c ⁻¹	\$8 kW _c ⁻¹ yr ⁻¹	\$0.50 MWh _c ⁻¹	25 yr
Li-ion BESS (G3)	\$405 kWh ⁻¹ + \$1,620 kW ⁻¹	\$25 kW ⁻¹ yr ⁻¹	\$0.50 MWh ⁻¹	15 yr
NGCC (Case 3)	\$1,330 kW _e ⁻¹	\$29.3 kW _e ⁻¹ yr ⁻¹	\$2.65 MWh _e ⁻¹	30 yr
ERCOT interconnection (Cases 0, 1, 2)	not capitalized	N/A	N/A	N/A

Technical parameters.

Asset	Other technical parameters
BWRX-300 reactor (Cases 1, 2)	870 MW _{th} , 270 MW _e net; $\eta_r = 0.345$; min-load fraction 0.50; ramp 1% min ⁻¹ ; capacity factor 0.92; fuel \$6 MWh _{th} ⁻¹ ; lifecycle emissions 12 g CO ₂ kWh _e ⁻¹
Cascaded steam turbine (Cases 1, 2)	$\eta_r = 0.345$ rated at zero extraction; auxiliary load 10%; Willans slope $\alpha_w = 0.083$ MW _e MW _{th} ⁻¹
Double-effect absorption chiller (Case 2)	160 MW _{th} nominal absorption unit; effective delivered cooling limited by steam input, COP and availability; nameplate cap COP 1.30; design $T_{wb,0} = 26$ °C, COP 1.10; de-rating 0.015 °C ⁻¹ ; crystallization $T_{cry} = 32$ °C cooling-water inlet; tower approach 5 K; parasitic $\beta_a = 0.020$ kW _e kW _c ⁻¹
Vapor-compression chiller (Cases 0, 1, 2, 3)	water-cooled centrifugal; nameplate COP 3.5; Houston-derated COP 3.2; 160 MW _c in all cases
Li-ion BESS (G3)	100 MWh, 50 MW (C/2); round-trip $\eta_{rt} = 0.85$; self-discharge 0.1% day ⁻¹ ; state-of-charge bounds [0.05, 0.95]
NGCC (Case 3)	200 MW _e F-class; HHV $\eta_{hhv} = 0.495$; direct CO ₂ 360 g kWh _e ⁻¹ ; upstream CH ₄ 60 g kWh _e ⁻¹ (CO ₂ -eq); Henry Hub + Houston Ship Channel basis \$0.10 MMBtu ⁻¹
ERCOT interconnection (Cases 0, 1, 2)	300 MW PCC bidirectional; flat substation interconnection in baseline

Table S4 Market inputs and on-site economics by operating year. ERCOT denotes Electric Reliability Council of Texas; LMP, locational marginal price; TAC, total annualized cost; NGCC, natural gas combined cycle. ERCOT LMP is the annual-mean Houston-zone day-ahead price; delivered gas is the annual-mean Henry Hub plus Houston Ship Channel basis. The grid-cost margin M is positive when the configuration is cheaper than grid procurement.

Quantity	Unit	2022	2023	2024
ERCOT day-ahead LMP	\$ MWh _e ⁻¹	70	57	28
Delivered natural gas	\$ MMBtu ⁻¹	6.53	2.63	2.35
Grid-only (Case 0) TAC	\$M yr ⁻¹	86.2	78.1	36.9
NGCC (Case 3) TAC	\$M yr ⁻¹	87.1	55.7	53.6
NGCC grid-cost margin M	%	-1.1	+28.7	-45.1

Table S5 Baseline secondary cost and carbon endpoints at the 2023 ATB-Mid baseline. LCOE denotes levelized cost of electricity (\$ MWh_e⁻¹); LCOC, levelized cost of cooling (\$ MWh_c⁻¹); EPBT, energy payback time (yr); CAC, carbon abatement cost (\$ tCO₂⁻¹). Net CO₂ is reported in kt yr⁻¹. CAC is reported only for the low-carbon nuclear configurations relative to the grid-only case. Cells marked -- are not defined for that configuration in the run-level KPI table.

Configuration	LCOE	LCOC	Net CO ₂	EPBT	CAC
Case 0 grid-only	94.1	25.3	+369	--	--
Case 1 SMR	231.6	--	-388	0.52	150.8
Case 2 SMR+abs.	234.5	--	-448	0.53	142.7
Case 3 NGCC	67.1	--	+486	0.18	--

Table S6 Baseline water-footprint endpoints at the 2023 ATB-Mid baseline. Direct site water includes data-center and cooling-system water use; indirect generation water accounts for the upstream electricity or on-site generation water factor. Direct, indirect and total water are reported in L MWh_e⁻¹; scarcity-weighted water is reported in m³ world-eq MWh_e⁻¹.

Configuration	Direct site	Indirect gen.	Total water	Scarcity-weighted
Case 0 grid-only	111	1,888	1,999	1.30
Case 1 SMR	111	8,688	8,799	5.72
Case 2 SMR+abs.	197	8,294	8,491	5.52
Case 3 NGCC	111	1,037	1,148	0.75

Table S7 Marginal annual value of adding the absorption chiller to a nuclear-only plant (Case 2 minus Case 1), at the 2023 ATB-Mid baseline (full-load PUE 1.35) and across the cooling-efficiency sweep. ATB-Mid denotes the mid-range National Laboratory of the Rockies Annual Technology Baseline reactor-capital case; PUE, power usage effectiveness; VCC, vapor-compression chiller; O&M, operations and maintenance. Positive values reduce Case 2 cost relative to Case 1; negative values increase it. All entries are \$M yr⁻¹.

Component	Baseline	PUE 1.10	PUE 1.30	PUE 1.50
Gross avoided VCC electricity	+18.24	+5.25	+15.76	+26.26
VCC backup and top-up	-2.26	-3.31	-1.95	-3.25
Absorption capital + fixed O&M	-13.22	-13.22	-13.22	-13.22
Willans extraction opportunity cost	-4.15	-1.46	-4.15	-4.15
Production tax credit forgone	-0.62	-0.03	-0.62	-0.62
Direct site-water cost	-0.04	-0.01	-0.04	-0.04
Residual reconciliation	-0.35	-0.18	-0.35	-0.35
Net value of absorption	-2.39	-12.96	-4.56	+4.64

Table S8 Data-center size-matching results for Case 2 (group G8) at the 2023 ATB-Mid baseline. ATB-Mid denotes the mid-range National Laboratory of the Rockies Annual Technology Baseline reactor-capital case; IT, information technology; TAC, total annualized cost. The reactor is held at 270 MW_e net while the IT load, cooling and NGCC capacities scale with the multiplier; the PCC remains 300 MW below 1× and scales upward above 1×. M is the grid-cost margin against the matched grid-only baseline; net export is annual grid sales minus purchases.

Load multiplier	0.5	1.0	1.5	2.0	2.5	3.0
Case 2 TAC (\$M yr ⁻¹)	154.3	194.6	234.8	275.0	315.3	355.5
Grid-only TAC (\$M yr ⁻¹)	39.0	78.1	117.1	156.1	195.1	234.2
Annual gap (\$M yr ⁻¹)	115.3	116.5	117.7	118.9	120.1	121.3
Grid-cost margin M (%)	-295	-149	-101	-76	-62	-52
Net grid export (TWh yr ⁻¹)	+1.89	+1.43	+0.97	+0.51	+0.04	-0.42
Cost intensity (\$ MWh _{IT} ⁻¹)	372	235	189	166	152	143
Excess over grid (\$ MWh _{IT} ⁻¹)	278	140	95	72	58	49

Table S9 Baseline and sensitivity scenario run ledger. C0–C3 denote Cases 0–3; PUE, power usage effectiveness; ATB-Mid, mid-range National Laboratory of the Rockies Annual Technology Baseline reactor-capital case; CAPEX, overnight capital expenditure; WACC, weighted-average cost of capital; ERCOT, Electric Reliability Council of Texas; BESS, battery energy storage system; SMR, small modular reactor; FOAK, first-of-a-kind; NOAK, nth-of-a-kind; IT, information technology. Cells marked -- indicate that the case is not included in that scenario group.

Group	Sweep	Levels	Runs	C0	C1	C2	C3
G0	Baseline	2023, full-load PUE 1.35, ATB-Mid CAPEX, WACC 6.7%	4	✓	✓	✓	✓
G1	Effective PUE	1.10, 1.30, 1.50	6	–	✓	✓	–
G2	ERCOT year	2022, 2023, 2024	12	✓	✓	✓	✓
G3	BESS	on, off	8	control	✓	✓	control
G4	SMR CAPEX	FOAK, ATB-Mid, NOAK	6	–	✓	✓	–
G5	SMR × Abs. CAPEX	5 × 5 grid	25	–	–	✓	–
G6	Carbon price	\$0, \$50 and \$100 tCO ₂ ^{−1}	12	✓	✓	✓	✓
G7	WACC	5, 6.7, 10%	3	–	–	✓	–
G8	Data-center size	0.5–3.0× IT load	24	✓	✓	✓	✓
Total			100				

Note: In G3, control denotes duplicated control rows retained in the run ledger; BESS dispatch is optimized only for Cases 1 and 2 because Cases 0 and 3 are closed-form annual cost evaluations.